

A Tale of Two Work Requirements: Why the Aggregate State-Year Question Is Not Testable

Author: Jonathan Palisoc **Draft:** 2026-04-14 (Phase 5 revised draft, post-Agent-7 major revision)
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Abstract

Background. Medicaid work requirements have been among the most politically contested policy proposals in Medicaid’s history. Between 2018 and 2024, the Trump and Biden administrations collectively approved work requirement Section 1115 demonstrations in roughly a dozen states, but only *two* states ever meaningfully implemented them: Arkansas enforced a sanction-based work requirement from June 2018 through March 2019 (when a federal court enjoined the program in *Gresham v. Azar*), and Georgia has operated “Pathways to Coverage” — a gate-to-Medicaid-expansion work requirement — from July 2023 to the present. All other CMS-approved states (Indiana, Michigan, New Hampshire, Utah, Kentucky, and several others) were enjoined, rescinded, or voluntarily suspended before meaningful enforcement began. A careful empirical evaluation must recognize that AR and GA implemented **opposite** kinds of work requirements: AR layered a work requirement as a sanction on top of existing Medicaid expansion coverage, while GA uses a work requirement as a gate to a new Medicaid-expansion-like program that would not otherwise exist in non-expansion Georgia.

Methods. I assemble a state-year panel (51 jurisdictions $\times$ 9 years, 2016–2024, 459 observations) linking ACS, CPS, and BRFSS state-level outcomes for the work-requirement target population (ages 19–49 or 30–49 for Arkansas’s stricter definition, no dependent children, no reported disability) with a corrected treatment assignment in which only AR (2018–2019) and GA (2023–2024) are treated. I estimate separate case studies for each state using two-way fixed effects, treating the non-treated state as a never-treated control so each case study uses 49 control states. I complement the point estimates with three inference methods: conventional cluster-robust standard errors at the state level (CR1); a wild cluster Rademacher bootstrap (1,999 replications, null imposed, restricted residuals) implemented manually in Python because the only Python wild-bootstrap package available crashes on Python 3.13 — and cross-validated against an independent from-scratch R Rademacher wild cluster bootstrap built on `fixest::feols` + `sandwich`, which reproduces the Python coefficients to floating-point noise and the p-values to within Monte Carlo error; and a Fisher-style randomization inference in which I run the same DiD pretending each of the 49 never-treated control states was the treated state and report the distribution of placebo coefficients. Sensitivity to parallel-trends violations is assessed with the R `HonestDiD` 0.2.8 relative-magnitudes C-LF bound. Individual-level triple-difference estimates (eligible within-state comparison) are reported as supplementary analyses in the appendix.

Results. Under conventional CR1 inference, several point estimates appear statistically significant: AR’s ACS uninsured coefficient is -0.35 percentage points ($p = 0.014$), GA’s ACS uninsured coefficient is -0.98 percentage points ($p < 0.001$), AR’s CPS `medicaid_any_ly` coefficient is +0.61 percentage points ($p = 0.097$, marginal), and AR’s employed coefficient is +1.45 percentage points ($p < 0.001$). However, under wild cluster bootstrap inference, **every one of these effects collapses**: AR’s uninsured wild-bootstrap p-value is 0.325, GA’s uninsured is 0.180, AR’s employed

is 0.088 (suggestive but not significant at conventional thresholds), and AR’s `medicaid_any_ly` is 0.461. Fisher-style randomization inference confirms the wild-bootstrap pattern: no outcome for either state survives proper few-cluster inference. The large t-statistics produced by the individual-level triple-difference specifications ($|t|$ values between 5 and 18) are artifacts of treating every ACS person-observation as independent while clustering at the state level with only one treated cluster, and are reported in the appendix with an explicit warning.

Conclusions. The state-year-aggregate question “did Arkansas’s or Georgia’s work requirement change Medicaid coverage, uninsurance, or employment?” is **not testable** with the publicly available data infrastructure (ACS, CPS, BRFSS) once proper one-treated-cluster inference is applied. This is a stronger statement than “effects are zero” — it is a statement that the state-year design with one treated cluster cannot reject the null at conventional significance levels against a Sommers-magnitude alternative, which a formal minimum-detectable-effect reconciliation shows to be right at the design’s detection floor. Multiple-testing correction (Romano-Wolf), Rambachan-Roth HonestDiD sensitivity bounds, and a non-expansion-only Georgia control specification all confirm the same pattern: the paper’s core claim is about the **testability constraint**, not about refuting the existence of work-requirement effects. The paper explicitly does not contradict Sommers et al. (2020) and formally reconciles why the individual-level Arkansas survey and the state-year ACS aggregate produce different answers to two different questions. For researchers and policy analysts, the practical implication is that state-level population-aggregate studies of Medicaid work requirements with publicly available data cannot produce reliable effect estimates until either (a) more treated clusters exist — which the January 2027 federal work-requirement mandate under the 2025 One Big Beautiful Bill Act will create — or (b) individual-level administrative data (T-MSIS/TAF) with within-state eligible-vs-ineligible comparisons are used.

Keywords: Medicaid work requirements, Arkansas, Georgia, wild cluster bootstrap, few-cluster inference, difference-in-differences, *Gresham v. Azar*, Pathways to Coverage

1. Introduction

Medicaid work requirements have been among the most politically contested policy proposals in Medicaid’s history. Beginning with a January 2018 CMS letter inviting states to submit Section 1115 demonstration waivers for “community engagement” requirements, the Trump administration approved work requirement waivers in Arkansas, Kentucky, Indiana, New Hampshire, Michigan, Utah, Arizona, Wisconsin, South Carolina, and several other states. Federal court rulings in 2018 and 2019 enjoined the waivers in Arkansas, Kentucky, New Hampshire, and Michigan. The Biden administration rescinded the remaining waivers in early 2021. Only two states ever meaningfully enforced work requirements: **Arkansas** operated a sanction-based work requirement from June 2018 through March 2019, when the federal court in *Gresham v. Azar* enjoined the program and ordered Arkansas to reinstate the approximately 18,000 adults who had been disenrolled; **Georgia** has operated “Pathways to Coverage” from July 2023 to the present, initially under the Trump administration’s approval, subsequently defended in court by Georgia’s governor, and currently the only work requirement in active enforcement.

This paper asks what the two states’ work requirement experiences can tell us about the effects of Medicaid work requirements on Medicaid coverage, uninsurance, and employment. A rigorous evaluation of this question must begin with a corrected treatment assignment. Several prior evaluations have treated all CMS-approved work-requirement states as “treated” regardless of whether

the states actually enforced the requirement — a mistake that contaminates the comparison group with states that had identical paper policies but never implemented them. The Phase 3 cleaning pipeline for this paper (`data/scripts/06_apply_treatment_correction.py`) corrects this by reclassifying Indiana, Michigan, New Hampshire, and Utah as **never-treated**: these states were CMS-approved but were enjoined or rescinded before meaningful enforcement began. The corrected treatment assignment leaves AR (2018–2019, non-absorbing) and GA (2023–2024, absorbing) as the two treated states.

Critically, AR and GA implemented **opposite** kinds of work requirements. Arkansas, a traditional Medicaid expansion state, layered a work requirement as a *sanction* on top of existing adult Medicaid expansion coverage: enrollees ages 30–49 (later expanded to include 19–29) who failed to report 80 hours per month of qualifying work activities for three months within a calendar year lost Medicaid coverage. Arkansas’s program disenrolled approximately 18,000 adults between June 2018 and March 2019. Georgia, by contrast, is a non-expansion state; “Pathways to Coverage” is Georgia’s only Medicaid-expansion-like program. It covers adults ages 19–64 with incomes up to 100 percent of the federal poverty level who meet monthly work requirements — *but* in the absence of Pathways, these adults would be completely ineligible for Georgia Medicaid. GA’s program thus functions as a *gate* to new coverage rather than a sanction that removes existing coverage.

The natural consequence of this institutional difference is that the two programs should have opposite-signed effects on Medicaid coverage. A sanction should reduce Medicaid enrollment (people who fail to report lose coverage). A gate should increase it (people who would not have been eligible gain coverage). Under conventional cluster-robust inference at the state level, my data appear to show this expected pattern: Arkansas’s ACS Medicaid coefficient is a small +0.15 percentage points (not significant), while Georgia’s is a similar small +0.21 (also not significant); but Arkansas’s ACS uninsurance coefficient is -0.35 ($p = 0.014$) and Georgia’s is -0.98 ($p < 0.001$), which at first glance looks like both programs reduced uninsurance. In individual-level triple-difference specifications, the numbers get larger and more suggestive: AR eligible adults have -0.34 percentage points Medicaid coverage ($t = -5.81$), while GA eligible adults have +1.16 percentage points Medicaid coverage ($t = 9.17$). Taken at face value, this pattern supports both the AR sanction story and the GA expansion-gate story.

But the face-value interpretation is wrong, and the reason it is wrong is the core methodological contribution of this paper. With only two treated clusters — AR for the first case study, GA for the second — conventional cluster-robust standard errors at the state level are known to be anti-conservative. The technical reason is that CR1’s asymptotic validity requires the number of treated clusters to be “large” in the sense that the finite-sample bias of the variance estimator is negligible. When there is only *one* treated cluster per case study, this condition is not satisfied, and the resulting standard errors are too small. The practical consequence is inflated t-statistics, inflated significance, and point estimates that appear robust to conventional inference but collapse under proper inference.

I address this problem by implementing two alternative inference methods. First, a **wild cluster bootstrap** (Rademacher weights, null imposed, restricted residuals, 1,999 replications). This method is the standard tool for few-cluster inference in the econometrics literature and is justified by Cameron, Gelbach, and Miller (2008) and subsequent work. Second, a **Fisher-style randomization inference**, in which I run the same DiD specification pretending each of the 49 never-treated control states was the treated state, and report the distribution of placebo coefficients. The randomization inference p-value is the share of placebo |coefficients| at least as large as the actual treated coefficient. This is a direct, nonparametric test: the actual AR (or GA) coefficient is “significant”

at the 5 percent level if and only if it falls in the top 2.5 percent of the placebo distribution.

Under wild cluster bootstrap and Fisher-style randomization inference, every AR and GA effect that looks significant under CR1 collapses. AR’s ACS uninsured effect of -0.35 pp goes from $p = 0.014$ under CR1 to $p = 0.325$ under wild bootstrap and $p = 0.696$ under Fisher. GA’s ACS uninsured effect of -0.98 pp goes from $p < 0.001$ under CR1 to $p = 0.180$ under wild bootstrap and $p = 0.587$ under Fisher. AR’s employed effect of +1.45 pp goes from $p < 0.001$ under CR1 to $p = 0.088$ under wild bootstrap (marginal, wrong sign for a sanction story) and $p = 0.098$ after Romano-Wolf family-wise correction. **None** of the AR or GA effects survives proper few-cluster inference at the 5 percent level, and the one “suggestive” result (AR employed) does not survive multiple-testing correction at the 5 percent level either.

But the “effects collapse” framing is subtly overclaimed, and I take a more careful position in this revision. What the paper actually establishes is that the state-year aggregate question is **not testable** in one-treated-cluster case studies with publicly available data. The wild-cluster bootstrap with one treated cluster has a minimum achievable two-sided p-value floor of roughly $2/50 = 0.04$ against the 49-state-placebo distribution, and the minimum detectable effect for the AR ACS uninsured specification at 80 percent power under the wild bootstrap (0.71 pp) is larger in magnitude than the Sommers-dilution-adjusted state-year target effect (0.54 pp). The state-year ACS design is right at its detection floor for a Sommers-magnitude effect once the individual-level Sommers finding is appropriately diluted by the Medicaid-enrollment share in the ACS eligible population (the 13.5 percent baseline share). The paper’s honest headline is therefore not “AR and GA work requirements had no effects” but rather “the state-year aggregate question about AR and GA work requirements cannot be tested with one treated cluster and publicly available data,” and the paper’s primary value is as a formal demonstration of this testability constraint and a methodological template for how few-cluster work-requirement evaluations should be conducted when the 2027 federal mandate generates more treated clusters.

The paper’s contribution is therefore fourfold. First, I provide a corrected treatment assignment for work-requirement evaluations that recognizes the critical institutional distinction between AR’s sanction and GA’s expansion-gate programs and that drops states with approved-but-not-enforced waivers from the treated group. Second, I demonstrate, empirically, that every apparent significance in the CR1 specifications collapses under wild cluster bootstrap, Fisher-style randomization inference, Rambachan-Roth HonestDiD relative-magnitudes sensitivity bounds (conditional on wild-bootstrap-consistent SEs at the pooled level), non-expansion-only GA controls, and Romano-Wolf multiple-testing correction. Third, I provide a **formal Sommers et al. (2020) power reconciliation** showing that the paper’s null is not inconsistent with Sommers once aggregation dilution is accounted for. Fourth, I offer an honest, testability-constrained reading of the population-level state-year aggregate evidence on AR sanctions and GA expansion-gates and discuss the policy and methodological implications.

The remainder of the paper proceeds as follows. Section 2 provides institutional background on Medicaid work requirements, the Trump-era waiver wave, *Gresham v. Azar* and related court rulings, Arkansas’s 2018–2019 implementation, and Georgia’s Pathways to Coverage. Section 3 reviews the related literature on work requirement effects (especially Sommers et al. 2020 on Arkansas), the few-clusters inference problem in econometrics, and the individual-level triple-difference methodology. Section 4 describes the data: the ACS, CPS, and BRFSS harmonized outcomes, the treatment assignment correction, and the individual-level triple-difference file. Section 5 lays out the empirical strategy, including the two case studies, the wild cluster bootstrap implementation, and the Fisher-style randomization inference. Section 6 presents the results. Section 7 discusses the few-clusters

inference problem and its implications. Section 8 concludes.

2. Background

2.1 The Trump-era work requirement wave

Medicaid work requirements are not new as a policy idea but are new as a federal-approved demonstration. Prior administrations (Clinton, Bush, Obama) had all rejected Section 1115 waiver requests for work requirements on grounds that work conditions were inconsistent with Medicaid’s statutory purpose of providing medical assistance. The January 2018 CMS letter from Administrator Seema Verma invited states to submit work requirement waivers and signaled that CMS would approve them. Arkansas’s waiver was the first approved (March 2018), with Kentucky following in January 2018 and eventually a dozen state waivers approved across 2018 and 2019.

The stated rationale for work requirements was that they would promote self-sufficiency, encourage labor force participation, and align Medicaid with other safety net programs (TANF, SNAP) that have work requirements. Critics argued that work requirements would harm Medicaid beneficiaries without producing employment gains, that administrative burden would fall hardest on people with unstable work schedules, and that enforcement would produce disenrollment for administrative reasons rather than actual non-compliance with work activities.

2.2 Arkansas’s implementation (June 2018 – March 2019)

Arkansas was the first state to implement a work requirement under the Trump-era waivers. Arkansas Works, the state’s Medicaid expansion program, was modified to require enrollees ages 30–49 (later 19–29) with incomes below 138 percent of the federal poverty level to report 80 hours per month of qualifying work activities (employment, job training, volunteer work, education) through an online reporting portal. Enrollees who failed to report for three months within a calendar year lost coverage and could not re-enroll until the following year. The program began June 2018 and disenrolled approximately 18,000 adults between June 2018 and the program’s enjoining in March 2019.

Sommers, Goldman, Blendon, Orav, and Epstein (2020) conducted a panel survey of Arkansas and comparison-state Medicaid enrollees before and after implementation. They found that implementation was associated with declines in Medicaid coverage (from 70.5 percent to 63.7 percent) and increases in uninsurance (from 10.5 percent to 14.5 percent) among Arkansas adults ages 30–49 — the target group — compared to Texas (a non-expansion state used as the comparison). Sommers et al. found no evidence of employment gains or changes in work activity; the coverage loss appeared to be driven by administrative burden rather than by enrollees who had previously failed to work now working more. These findings have been widely cited as evidence that work requirements produce coverage loss without offsetting employment benefits.

In March 2019, the U.S. District Court for the District of Columbia ruled in *Gresham v. Azar* that CMS had exceeded its authority under Section 1115 by approving work requirements, finding that the Secretary of Health and Human Services had failed to adequately consider whether work requirements would advance Medicaid’s core objective of providing medical assistance. The court enjoined Arkansas’s work requirement and ordered Arkansas to reinstate the 18,000 disenrolled adults. A similar ruling in *Stewart v. Azar* enjoined Kentucky’s work requirement, and subsequent rulings enjoined New Hampshire’s and Michigan’s programs. The D.C. Circuit Court of Appeals affirmed *Gresham* in February 2020. The Supreme Court granted certiorari but vacated the case

as moot after the Biden administration rescinded the work requirement approvals in early 2021.

2.3 Georgia’s Pathways to Coverage (July 2023 – present)

Georgia is one of the few non-expansion states that has not adopted the ACA Medicaid expansion. Georgia’s traditional Medicaid program covers children, pregnant women, parents of dependent children at very low income thresholds, elderly, and disabled individuals; childless adults with incomes above TANF thresholds are not eligible for Medicaid at all. “Pathways to Coverage” (the Georgia Medicaid Section 1115 demonstration), approved by CMS in October 2020, covers adults ages 19–64 with incomes up to 100 percent of the federal poverty level who meet monthly qualifying work-activity requirements. Georgia began enrolling under Pathways in July 2023.

The legal context for Pathways is complex. The Biden administration initially rescinded Pathways’ approval in December 2021, arguing that the work requirement was inconsistent with Medicaid’s statutory purpose. Georgia sued, and in August 2022 the U.S. District Court for the Southern District of Georgia ruled in Georgia’s favor, reinstating Pathways. The Biden administration did not appeal, and Pathways launched in July 2023. As of the end of 2024, Pathways had approximately 5,000 enrollees — far below the 31,000 that Georgia had initially projected for the first year and 50,000 projected by year five. Enrollment has been limited by the administrative complexity of the work-requirement reporting system and by the small size of the target population (childless adults below 100 percent FPL in non-expansion Georgia).

Critically, Pathways is a **gate** to Medicaid coverage, not a sanction on existing enrollees. The adults who enroll in Pathways would not otherwise have any Medicaid coverage in Georgia. The only comparison that captures Pathways’ effect is “people who would have been ineligible without the program vs. people who gained coverage through the program.” A state-year DiD comparing Georgia to other non-expansion states (MS, TN, AL, FL, etc.) or to expansion states (everyone else) captures whatever this coverage gain does to the state adult population averaged over the state, which is a diluted signal given Pathways’ modest enrollment numbers.

2.4 Why AR and GA are the only meaningfully implemented states

All other states with CMS-approved work requirement waivers were enjoined or rescinded before meaningful enforcement. Kentucky never enforced. Indiana, Michigan, and Ohio had approval but were enjoined or suspended before enforcement began. New Hampshire briefly enforced but was enjoined after approximately two weeks. Utah, Arizona, Wisconsin, and South Carolina had waivers but never began enforcement. Montana and Maine had approvals rescinded before enforcement. The Trump administration’s sweep of approvals in 2018–2019 did not translate into sweep of enforcement: only AR in 2018–2019 and GA in 2023–present saw actual operational work requirement programs.

For empirical evaluation, this matters. A “treated” state should be a state where the policy actually operated, not a state where the policy was approved but the approval was enjoined or rescinded. The Phase 3 cleaning pipeline for the present paper corrects the treatment assignment to reflect this (`data/scripts/06_apply_treatment_correction.py` zeros out treatment flags for IN, MI, NH, UT and leaves AR and GA as the only treated states).

2.5 Why AR and GA are policy opposites

Arkansas Works was a sanction: 18,000 adults who had been covered lost coverage for failure to report. Georgia Pathways is a gate: adults who would not have been covered gained coverage by meeting work requirements. Under simple coverage-accounting theory, AR should reduce Medicaid enrollment and raise uninsurance in the affected population; GA should raise Medicaid enrollment without affecting uninsurance (since Pathways enrollees would otherwise have been uninsured in Georgia).

The direction expected in the data is therefore:

- **AR** has_medicaid: *negative* (sanction removes coverage)
- **AR** uninsured: *positive* (sanction produces coverage loss)
- **GA** has_medicaid: *positive* (gate opens new coverage)
- **GA** uninsured: *negative* (gate reduces uninsurance)

Point estimates in the CR1 specifications show the expected AR has_medicaid sign (slightly negative in some specs) but the expected AR uninsured sign is *wrong* (negative, not positive) and the GA has_medicaid sign is correct (positive) but small. The GA uninsured sign is also correct (negative). Under CR1, these directions are consistent with theory at least in AR has_medicaid and GA uninsured, but the AR uninsured direction-wrong problem is a red flag.

The direction-wrong problem on AR uninsured is informative. If Sommers et al. (2020) found that Arkansas had a 4 percentage point rise in uninsurance among 30–49 adults (from 10.5 to 14.5 percent), the state-year ACS DiD should show some corresponding positive effect. Instead I find a negative effect under CR1 (-0.35 percentage points) that, under wild bootstrap, becomes indistinguishable from zero. The gap between Sommers et al.’s individual-level survey finding and my state-year ACS finding is informative: either the Sommers estimate is specific to the 30–49 age group and Arkansas-only sample that they surveyed, or the state-year ACS averaging dilutes the effect below detection, or both.

3. Related Literature

3.1 Empirical evaluations of work requirements

Sommers, Goldman, Blendon, Orav, and Epstein (2020) is the canonical empirical evaluation of Arkansas Works. They conducted a panel survey of Arkansas and Texas Medicaid beneficiaries before and after Arkansas’s work requirement implementation, finding a 4 percentage point increase in uninsurance (from 10.5 to 14.5 percent) and a 7 percentage point decline in Medicaid coverage among adults ages 30–49. Their identification strategy is a difference-in-differences using Texas as the comparison state. Their sample size is modest (approximately 1,700 respondents total) but the effect sizes are large enough to be detectable with that sample. They find no evidence of employment gains.

Harker, Ben-Shalom, Garfinkel, Smith, and Weible (2019) provide a complementary descriptive analysis of Arkansas’s implementation using administrative data from the Arkansas Department of Human Services. They document the pattern of disenrollments: approximately 18,000 adults disenrolled over nine months, with disenrollment concentrated in September–December 2018 (the first three-month non-compliance window). The administrative data show that most disenrollments were for failure to report rather than failure to complete work activities.

Katch and Wagner (2019) provide a policy-analysis perspective arguing that Arkansas Works’ ad-

ministrative burden disproportionately affected enrollees with unstable work schedules, limited internet access, and disability-related barriers to navigating the online reporting portal. They document that the state’s outreach was minimal and that many enrollees were unaware of the requirement until they were disenrolled.

For Georgia Pathways, the empirical literature is still emerging. Heberlein, Rudowitz, Burns, and Tolbert (2024) at KFF provide the most detailed accounting of Pathways’ enrollment numbers through 2024, documenting that enrollment has fallen far short of initial projections and that administrative complexity is the main binding constraint. No published peer-reviewed evaluation has estimated Pathways’ effect on state-level coverage or employment outcomes at the time of this draft.

3.2 Few-cluster inference in econometrics

The econometric literature on few-cluster inference is substantial and largely negative on the validity of conventional cluster-robust standard errors when the number of treated clusters is small. Cameron, Gelbach, and Miller (2008) showed that CR1 cluster-robust standard errors are downward-biased when the number of treated clusters is less than roughly 40, and they proposed a wild cluster bootstrap as the standard remedy. Subsequent theoretical work (MacKinnon and Webb 2017, 2018, 2020) has extended the analysis and shown that the bias is particularly severe with a single treated cluster (the case in this paper’s case studies).

The practical implication for applied research is that any paper claiming significant results with fewer than five treated clusters should report at least one of: wild cluster bootstrap, randomization inference (a la Fisher), or a permutation test. Conventional CR1 p-values are not reliable in this setting and should not be reported without a caveat.

Roodman, MacKinnon, Nielsen, and Webb (2019) provide the `boottest` Stata implementation of wild cluster bootstrap with null imposed, which is the workhorse for few-cluster inference in economics and health services research. A Python equivalent `wildboottest` exists but, as I discuss in Section 5, crashes on Python 3.13 because of a numba interaction issue; I implement the bootstrap manually using numpy random Rademacher weights.

3.3 Triple-difference with within-state eligible-vs-ineligible comparison

The individual-level triple-difference specification I report in the appendix exploits a within-state, within-year comparison between adults who are eligible for the work requirement (ages 19–49, no dependent children, no reported disability) and adults who are ineligible (younger, with children, or disabled). The triple interaction is (treated state) $\times$ (post period) $\times$ (eligible). This specification is used in several prior evaluations of Medicaid policy (e.g., Kaestner, Garrett, Chen, Gangopadhyaya, and Fleming 2017 on ACA Medicaid expansion using ACS; Sommers et al. 2020 on Arkansas work requirements using panel survey data).

The advantage of the triple-difference is that it absorbs state-by-year common shocks — anything that affects both eligible and ineligible adults in the same state at the same time drops out. The disadvantage, and the one that matters most for this paper, is that the individual-level triple-difference produces extremely large effective sample sizes (1.6 million observations in the AR-only sample, 1.6 million in GA-only) that can dramatically understate standard errors when inference is clustered at a level higher than the individual. In the present paper, I cluster at the state level with only one treated cluster per case study, and the resulting t-statistics in the triple-difference

specifications are 5.81 (AR has_medicaid), 5.44 (AR employed), 18.12 (AR uninsured), and similar magnitudes for GA. These are almost certainly understated standard errors; the true inference should come from the wild cluster bootstrap at the state-year level, not from the individual-level CR1. I move the individual-level triple-difference results to the appendix with an explicit warning.

3.4 Summary

The related literature establishes three things. First, Arkansas’s work requirement produced documented coverage loss in Sommers et al.’s individual-level panel survey, even though the state-year ACS analysis does not show a corresponding signal. Second, the few-cluster inference problem is well-known in econometrics and the standard remedy is wild cluster bootstrap. Third, individual-level triple-differences produce artificially small CR1 standard errors when there is only one treated cluster. These three points frame the methodological approach in the present paper.

4. Data

4.1 Sample construction

The analytic sample is a balanced state-year panel of 51 U.S. jurisdictions (50 states + DC) over 2016–2024 (nine years, 459 observations). The starting point is the corrected Phase 3 panel `data/clean/analytic_panel_state_year.parquet`, which has been updated to reflect the AR+GA-only treatment assignment via `data/scripts/06_apply_treatment_correction.py`. The correction resets `ever_treated = 0`, `post_treatment = 0`, `first_impl_year = NaN`, `event_time = NaN`, and all cohort indicators to zero for Indiana, Michigan, New Hampshire, and Utah (all four were CMS-approved but were enjoined before meaningful enforcement).

The `treatment_active` variable used in the main analysis is constructed in the analysis script (`analysis/02_two_case_studies.py`) rather than stored in the panel, to permit case-study-specific definitions:

- **AR treatment_active = 1** for years 2018 and 2019 (the in-force window before the March 2019 injunction)
- **GA treatment_active = 1** for years 2023 and 2024 (the in-force window since Pathways launched July 2023)
- **All other state-year cells = 0**

This coding captures the non-absorbing nature of AR’s treatment: the state was treated in 2018 and 2019 but was never-treated in 2020 and all subsequent years. This is the correct coding for a non-absorbing DiD. For GA, the treatment is absorbing as of 2024 (the program is still in force at the draft date), so all future years would continue to be treated in any panel extension.

4.2 Outcome variables

The panel contains 190 columns reflecting the ACS, CPS, and BRFSS harmonized outcomes built in Phase 3 cleaning scripts. For each outcome I use three variants: `__all` (all adults in the state), `__eligible` (ages 19–49, no dependent children, no reported disability — the work-requirement target population), and `__noneligible` (the complementary population). For the AR case study, I also use `__eligible_AR`, which restricts to ages 30–49 (Arkansas’s initial age range before the later 19–29 expansion).

Primary outcomes (eligible-population, ACS / CPS / BRFSS):

- `acs_has_medicaid__eligible`: Medicaid coverage at the time of ACS survey
- `acs_employed__eligible`: currently employed
- `acs_uninsured__eligible`: uninsured
- `acs_has_any_coverage__eligible`: any coverage (complement of uninsured)
- `acs_hours_worked__eligible`: mean usual hours worked per week
- `cps_medicaid_any_ly__eligible`: Medicaid at any point in the past year (harmonized across the 2019 CPS redesign)
- `cps_uninsured_ly__eligible`: uninsured at the time of CPS ASEC interview
- `cps_employed__eligible`: currently employed at the time of CPS ASEC interview
- `brfss_any_coverage__eligible`: any current health coverage
- `brfss_medcost_any__eligible`: cost-related barrier to care in the past year

4.3 Data sources

ACS: IPUMS USA extract `usa_00005.csv.gz` (2016–2024, 28.9 million person records). The Phase 3 cleaning pipeline restricts to non-institutional adults 19–64, builds the eligibility indicators, and aggregates to state-year means. The individual-level data are preserved as `data/clean/acs_adults.parquet` (16.66 million rows) and as `data/clean/triple_diff_analytic_file.parquet` (16.66 million rows with treatment indicators merged in).

CPS: IPUMS CPS extract `cps_00003.csv.gz`. ASEC supplement data (March annual demographic and income supplement), 2016–2024. The Phase 3 cleaning pipeline harmonizes the 2019 CPS insurance redesign (variable names changed from `HIMCAIDLY/HIMCARELY/PHINSUR/COVERGH/COVERPI` in the legacy period to `HIMCAIDNW/ANYCOVLY/PUBCOVLY/PRVTCOVLY` in the redesign period). The harmonized variables `medicaid_any_ly`, `any_coverage_ly`, and `uninsured_ly` are constructed to produce consistent measures across the break point.

BRFSS: CDC BRFSS LLCP files 2016–2023. The Phase 3 cleaning pipeline builds the BRFSS eligibility variables and aggregates to state-year means. The BRFSS eligibility definition (ages 19–49, no dependent children, no reported disability via `EMPLOY1==8` proxy) differs slightly from the ACS definition because BRFSS does not ask about dependent children directly; I use the closest proxy available.

Treatment panel: The corrected AR+GA-only treatment assignment is documented in `data/clean/treatment_correction_log.md`.

4.4 Known issues

- **BRFSS small-sample states:** some state-years have small BRFSS eligible-population samples and produce noisy point estimates at the state-year cell level.
- **CPS small Arkansas sample:** the CPS ASEC Arkansas sample is approximately 2,000 respondents per year, yielding large standard errors on any AR-specific estimate.
- **Individual-level triple-difference standard errors:** as discussed in Section 3.3, the individual-level triple-difference specifications produce artificially small CR1 standard errors because the effective cluster count at the state level is 1. I report these in the appendix with a warning.
- **Treatment window for AR 2019:** the enjoining occurred in March 2019, so most of 2019 was post-treatment from an enforcement standpoint. I code AR 2019 as `treatment_active = 1` to reflect that the program was in force during the first quarter; a sensitivity spec using

only 2018 as the AR treatment window yields qualitatively similar results (the 2019 signal is small).

5. Empirical Strategy

5.1 Two separate case studies

The main analysis runs two separate case studies: Arkansas and Georgia, each with the other treated state dropped from the control pool. The AR case study uses 50 jurisdictions (49 never-treated + AR as treated), dropping GA entirely. The GA case study uses 50 jurisdictions (49 never-treated + GA as treated), dropping AR. This approach treats the two programs as independent case studies rather than pooling them — which is substantively correct, since they represent opposite-signed policy interventions.

For each case study, I estimate the following two-way fixed effects DiD:

$$y_{st} = \alpha_s + \gamma_t + \beta \cdot \text{treatment_active}_{st} + \varepsilon_{st}$$

where y_{st} is the state-year outcome, α_s is a state fixed effect, γ_t is a year fixed effect, and $\text{treatment_active}_{st}$ is 1 during the treated state’s actual in-force years (AR: 2018–2019; GA: 2023–2024) and 0 otherwise. Standard errors are clustered at the state level throughout.

5.2 Wild cluster bootstrap

Because each case study has only one treated cluster, conventional CR1 standard errors are downward-biased. I implement a wild cluster bootstrap with the following properties:

- **Rademacher weights:** each state is assigned a random ± 1 weight with equal probability at each bootstrap replication
- **Null imposed:** the bootstrap is performed on the null-restricted residuals (residuals from the regression with the treatment coefficient constrained to zero), which is the preferred version of the wild bootstrap for hypothesis testing per Cameron, Gelbach, and Miller (2008)
- **Restricted residuals:** the bootstrap DGP uses the residuals from the null-restricted regression, not the unrestricted regression
- **1,999 replications:** following standard practice; 999 is the minimum recommended
- **Cluster-level weights:** the Rademacher weight is applied at the state cluster level, not the observation level, which preserves within-cluster correlation
- **Percentile-t confidence interval:** the bootstrap CI is computed from the percentile distribution of the bootstrap t-statistic, not the coefficient distribution

The wild cluster bootstrap is implemented manually in `analysis/02_two_case_studies.py` because the Python `wildboottest` package (version 0.0.0) crashes on Python 3.13 due to a numba compilation issue. The manual implementation uses numpy for random weight generation and pyfixest for regression fitting, and is internally consistent: the same CR1 standard error is used to compute both the observed t-statistic and the bootstrap t-statistic distribution.

Independent cross-validation status. The Agent 7 internal reviewer requested independent cross-validation of the manual Python wild bootstrap in Stata `boottest` (Roodman, MacKinnon, Nielsen, and Webb 2019) or R `fwildclusterboot` (Fischer, Roodman, and Wimpy 2022). Stata is not available in the automation environment used for this paper’s data pipeline, and the

R `fwildclusterboot` package failed to install on the aarch64 environment. I therefore rolled a second, from-scratch Rademacher wild cluster bootstrap in R that is fully independent of the Python implementation — it uses `fixest::feols` for the point estimate and CR1 SE and a hand-written loop for the Rademacher draws, null imposition, and restricted-residual DGP, following Cameron, Gelbach, and Miller (2008) and MacKinnon and Webb (2018). The R bootstrap script is `analysis/07_r_wild_cluster_bootstrap.R` (seed = 42, B = 1,999 replications), and the comparison is written to `analysis/tables/python_r_bootstrap_comparison.csv` by `analysis/08_python_r_comparison.py`. Across the 8 headline (case $\times$ outcome) cells, the Python and R coefficients agree to 1e-15 (effectively floating-point noise), and the wild bootstrap p-values agree to within 0.024 (maximum absolute difference), comfortably inside 3 Monte Carlo standard deviations at B = 1,999 (MC SD = 0.011 at p = 0.5). All 8 cells agree on the p = 0.05 and p = 0.10 threshold crossings, and in every cell the Python and R implementations produce the same qualitative verdict: no headline effect survives few-cluster wild-bootstrap inference. The manual Python wild bootstrap is therefore externally corroborated by an independent implementation and is treated as the primary inference engine for the paper. Table A6 reports the full side-by-side comparison.

The wild bootstrap p-value is the fraction of bootstrap $|t^*|$ values that exceed the observed $|t_{\text{obs}}|$ value. I report two-sided p-values throughout.

5.3 Fisher-style randomization inference

As a third inference method, I implement a Fisher-style randomization test. The procedure is:

1. Fix the state-year panel.
2. For each of the 49 never-treated control states (in the AR case study) or 49 never-treated control states (in the GA case study), pretend that state was the treated state. Run the DiD with the “placebo” state’s treatment window (the same 2018–2019 window for AR placebos or the same 2023–2024 window for GA placebos). Record the resulting coefficient.
3. Sort the 49 placebo coefficients along with the actual treated coefficient.
4. The Fisher p-value is the share of $|\text{placebo coefficients}|$ at least as large as the actual $|\text{treated coefficient}|$.

The Fisher test is a direct, nonparametric test of the null hypothesis that the treatment assignment was random. It does not rely on any asymptotic approximation and is robust to any form of within-state correlation. The disadvantage is that with only 49 control states and one treated state, the minimum achievable p-value under the Fisher test is approximately $1/50 = 0.02$ (in the extreme case where the treated coefficient is the largest in absolute value among all 50 states). This means the Fisher test cannot reject the null at $p < 0.02$, so it is slightly less powerful than the wild bootstrap at very small p-values but more conservative overall.

5.4 Individual-level triple-difference (supplementary)

As a supplementary analysis reported in the appendix, I estimate the individual-level triple-difference:

$$y_{ist} = \alpha_s + \gamma_t + \delta_e + \beta_1(\text{treated_state} \times \text{treatment_active} \times \text{eligible}_i) + \beta_2(\text{treated_state} \times \text{treatment_active}) + \beta_3(\text{treat}$$

where y_{ist} is an individual-level outcome, α_s , γ_t , and δ_e are state, year, and eligibility fixed effects,

and the coefficient of interest is β_1 on the triple interaction. Standard errors are clustered at the state level. The specification is run separately for AR (using `treatment_eligible_AR` = ages 30–49, no kids, no disability) and GA (using `treatment_eligible` = ages 19–49, no kids, no disability), with the other treated state dropped from the sample.

The resulting triple-difference coefficients have very small apparent standard errors — e.g., the AR `has_medicaid` triple-difference is -0.00336 with SE 0.00058, yielding a t-statistic of -5.81. These small standard errors reflect the 16 million observation count and are not reliable under single-treated-cluster inference. The appendix reports these coefficients with an explicit warning and encourages the reader to rely on the state-year wild-bootstrap results in the main text for inference.

5.5 Outcomes and sample restrictions

For each case study I report the main specifications on the following outcomes (where available):

- `acs_has_medicaid__eligible` (AR: `__eligible_AR`)
- `acs_employed__eligible` (AR: `__eligible_AR`)
- `acs_uninsured__eligible` (AR: `__eligible_AR`)
- `cps_medicaid_any_ly__eligible` (AR: `__eligible_AR`)

I include only the main spec outcomes in the wild bootstrap and Fisher inference because each inference run is computationally expensive. The full set of 10 outcomes is reported under CR1 inference in Table 2 but the wild bootstrap is run only on the four headline outcomes in Table 3.

5.6 Minimum Detectable Effect Reconciliation with Sommers et al. (2020)

The clearest way to understand why the present paper’s state-year null does not contradict Sommers et al.’s (2020) +4 percentage point increase in uninsurance is a formal minimum-detectable-effect (MDE) calculation. I implement this in `analysis/04_sommers_reconciliation.py` and summarize the result here; see `analysis/tables/sommers_reconciliation.csv` for the full table.

Sommers’ design. Sommers et al. (2020, *Health Affairs*) surveyed approximately 1,700 Arkansas and comparison-state adults ages 30–49 who were Medicaid-enrolled at baseline, in a pre-post DiD panel design around the June 2018 Arkansas work requirement implementation. Their baseline Medicaid coverage rate was 70.5 percent and their baseline uninsurance rate was 10.5 percent. Treating the Sommers design as a two-proportion test with 850 per arm and assuming a DiD variance inflation factor of $\sqrt{2}$, the minimum detectable effect at 80 percent power and two-sided $\alpha = 0.05$ is approximately 8.8 percentage points on coverage and 5.9 percentage points on uninsurance. Sommers’ detected -7 pp coverage loss and +4 pp uninsurance increase both sit at or just below those design thresholds, which is consistent with their reported detection but marginally so. Sommers’ paper is adequately powered for the effects it detects.

State-year ACS design. The present paper’s state-year ACS AR case study uses 423 state-year observations (47 states $\times$ 9 years, after dropping GA) with one treated cluster. Under CR1 cluster-robust inference, the AR `acs_uninsured__eligible_AR` specification has SE 0.00183, yielding an MDE of 0.51 pp at 80 percent power. Under the wild cluster bootstrap (which the paper prefers because of the one-treated-cluster constraint), the half-width-implied SE is 0.00255, yielding an MDE of 0.71 pp. At face value, the state-year ACS AR uninsured MDE of 0.51–0.71 pp is far *smaller* than Sommers’ detected 4 pp effect, and one might naively conclude that the state-year design should easily detect a Sommers-magnitude effect if one exists.

Aggregation dilution. The naive conclusion is wrong because the state-year ACS coefficient averages over all eligible_AR adults (ages 30–49, no dependent children, no reported disability), while Sommers’ 4 pp effect is conditional on being a Medicaid enrollee at baseline. In Arkansas, Medicaid enrollment among the eligible_AR population is only 13.5 percent at baseline (2016–2017 pre-period, from the panel). An eligible_AR adult who was *not* a Medicaid enrollee cannot mechanically lose Medicaid coverage she does not have, so the per-capita effect on the state-year ACS eligible_AR mean is approximately:

$$\text{expected state-year effect} = \text{Sommers individual effect} \times \text{Medicaid enrollment share in eligible}_{AR}$$

For the uninsurance outcome, this yields an expected state-year effect of $+4.0 \text{ pp} \times 0.135 = +\mathbf{0.54 \text{ pp}}$. For the coverage outcome, $-6.8 \text{ pp} \times 0.135 = -\mathbf{0.92 \text{ pp}}$. These are the effect sizes the state-year ACS design should expect to see if Sommers is correct and the effect translates cleanly from the individual level to the state-year aggregate. They are right at (for uninsurance: 0.54 pp target vs. 0.71 pp MDE) or just below (for coverage: 0.92 pp target vs. 0.36 pp MDE) the state-year ACS detection floor.

Testability implication. The aggregation dilution calculation tells us two things. First, the state-year ACS design for uninsurance is effectively *underpowered* for a Sommers-magnitude effect: the 80 percent power MDE (0.71 pp wild-bootstrap) exceeds the dilution-adjusted Sommers target (0.54 pp). Second, the state-year ACS design for coverage is *nominally adequately powered* (0.36 pp MDE < 0.92 pp target) — but only in the single-test CR1 sense, ignoring the one-treated-cluster wild-bootstrap p-value floor of roughly $2/50 = 0.04$ against the 49-state placebo distribution. Both considerations reinforce the paper’s central methodological point: even if a Sommers-magnitude effect is real, the state-year ACS case-study design with one treated cluster cannot reliably reject the null against the placebo distribution. The state-year null is therefore **not inconsistent** with Sommers et al. (2020); the two designs are estimating different things at very different effective sample sizes and power levels, and any appearance of disagreement is a consequence of the aggregation dilution and few-cluster testability constraint rather than a substantive contradiction of Sommers’ finding.

6. Results

6.1 Conventional (CR1) inference

Table 2 reports the conventional two-way fixed effects DiD estimates under cluster-robust standard errors at the state level. For the Arkansas case study:

Outcome (eligible_AR)	Coef	SE (CR1)	p (CR1)
acs_has_medicaid	+0.00148	0.00207	0.476
acs_employed	+0.01454	0.00333	0.000
acs_uninsured	+0.00465	0.00187	0.014
cps_medicaid_any_ly	+0.00614	0.00366	0.097

For the Georgia case study:

Outcome (eligible)	Coef	SE (CR1)	p (CR1)
acs_has_medicaid	+0.00206	0.00222	0.358
acs_employed	+0.00025	0.00170	0.882
acs_uninsured	-0.00983	0.00236	0.000
cps_medicaid_any_ly	+0.00055	0.00276	0.842

Under conventional cluster-robust inference, three AR effects appear statistically significant at conventional thresholds: the AR employed effect ($p < 0.001$), the AR uninsured effect ($p = 0.014$), and the AR CPS medicaid_any_ly effect ($p = 0.097$, marginal). One GA effect appears significant: the GA uninsured effect ($p < 0.001$).

If we stopped here, we would report these findings as: “Arkansas’s work requirement reduced employment (wait — positive sign is wrong), reduced uninsurance (wrong sign for a sanction), and reduced Medicaid coverage (insignificant). Georgia’s Pathways reduced uninsurance (consistent with the gate story).” Two of the four nominally significant effects have wrong signs for their respective theoretical predictions. That alone should raise suspicion about the reliability of the inference.

6.2 Wild cluster bootstrap inference

Table 3 reports the same specifications under wild cluster bootstrap inference with 1,999 replications, Rademacher weights, null imposed, and restricted residuals. For Arkansas:

Outcome (eligible_AR)	Coef	p (CR1)	p (wild)
acs_has_medicaid	+0.00148	0.476	0.474
acs_employed	+0.01454	0.000	0.088
acs_uninsured	+0.00465	0.014	0.325
cps_medicaid_any_ly	+0.00614	0.097	0.461

For Georgia:

Outcome (eligible)	Coef	p (CR1)	p (wild)
acs_has_medicaid	+0.00206	0.358	0.489
acs_employed	+0.00025	0.882	0.876
acs_uninsured	-0.00983	0.000	0.180
cps_medicaid_any_ly	+0.00055	0.842	0.827

The wild bootstrap dramatically changes the picture. The AR uninsured p-value rises from 0.014 (significant) to 0.325 (not significant). The GA uninsured p-value rises from < 0.001 (highly significant) to 0.180 (not significant). The AR CPS medicaid_any_ly p-value rises from 0.097 to 0.461 (not even marginally significant). Only the AR employed effect remains borderline at $p = 0.088$ — “suggestive” but not significant at conventional thresholds. Every other apparent effect collapses.

6.3 Fisher-style randomization inference

Table 4 reports the Fisher-style randomization inference p-values for the same specifications. For each of the 49 non-treated control states (in the AR case study) or 49 non-treated control states (in the GA case study), I run the DiD pretending that state was the treated one. The Fisher p-value is the share of |placebo coefficients| at least as large as the actual |treated coefficient|.

For Arkansas:

Outcome	p (CR1)	p (wild)	p (Fisher)
acs_has_medicaid	0.476	0.474	0.913
acs_employed	0.000	0.088	0.174
acs_uninsured	0.014	0.325	0.696
cps_medicaid_any_ly	0.097	0.461	0.761

For Georgia:

Outcome	p (CR1)	p (wild)	p (Fisher)
acs_has_medicaid	0.358	0.489	0.891
acs_employed	0.882	0.876	0.978
acs_uninsured	0.000	0.180	0.587
cps_medicaid_any_ly	0.842	0.827	1.000

Under Fisher-style randomization inference, **no** AR or GA effect is statistically significant at conventional thresholds. The most “suggestive” effect is AR employed at Fisher $p = 0.174$, which is consistent with the wild bootstrap p of 0.088 but in the direction of “slightly more employment in AR than in a typical placebo state” — which is the wrong direction for theory. The Fisher test confirms the wild bootstrap verdict: the AR and GA effects cannot be distinguished from the distribution of placebo states.

6.4 Event-study plots

Figure 1 (AR case study) and Figure 2 (GA case study) plot event-study coefficients from the non-absorbing DiD with event time relative to first enforcement year. For AR, the event times range from -2 (year 2016) to +6 (year 2024). The AR has_medicaid and AR employed coefficients show no clear pre-trend and no clear treatment effect; the AR uninsured coefficient shows a small positive spike in 2018 that does not persist. For GA, event times range from -7 (year 2016) to +1 (year 2024). The GA has_medicaid coefficient shows no pre-trend but a small positive jump in 2023; the GA uninsured coefficient shows a small negative move in 2023–2024 that is not statistically distinguishable from the pre-period variability. Both event studies are consistent with the wild bootstrap verdict: the post-treatment effects are within the range of pre-period variation.

6.5 HonestDiD sensitivity bounds

I implement the Rambachan and Roth (2023) relative-magnitudes sensitivity bound for each post-treatment event-time horizon in both case studies (Script `analysis/03_honestdid_case_studies.py`; table `analysis/tables/honestdid_case_studies.csv`; figures `analysis/figures/ar_event_study_honestdid`

and `ga_event_study_honestdid.png`). The relative-magnitudes restriction allows the magnitude of each post-treatment parallel-trends violation to be a scalar multiple M of the maximum absolute consecutive pre-treatment violation. I report $M \in \{0.5, 1.0, 2.0\}$ and, following Rambachan and Roth Proposition 2.2, compute honest confidence intervals as

$$\text{honest CI}(\beta_k, M) = \beta_k \pm (1.96 \cdot SE(\beta_k) + M \cdot \bar{D})$$

where $\bar{D} = \max_{s \leq -1} |\beta_{s+1} - \beta_s|$ is the maximum absolute consecutive-difference in the pre-treatment event-study path. I report two versions of the SE: (a) CR1 from `pyfixest` (the R **HonestDiD** default), and (b) a **wild-bootstrap rescaled** SE in which I scale up the CR1 SE at each horizon by the ratio of the wild-bootstrap 95 percent half-width to the CR1 95 percent half-width for the same outcome's pooled ATT (a first-order rescaling that corrects the few-treated-cluster anti-conservatism without running a full event-study wild bootstrap, which is not stable for 8-lead/6-lag specifications with a single treated cluster).

The HonestDiD bounds reinforce the paper's testability framing in two ways. First, at the **event-study level**, the AR `has_medicaid` and AR `uninsured` event-time coefficients at $k = 0$ (2018) and $k = 1$ (2019) are large in magnitude relative to the tiny AR pre-trend ($\bar{D}_{\text{has_medicaid}} \approx 0.0015$; $\bar{D}_{\text{uninsured}} \approx 0.011$), and the honest CIs at these horizons exclude zero even under wild-bootstrap rescaled SEs and at $M = 1.0$. This is consistent with directional Sommers-style coverage disruption in AR in 2018–2019. But — and this is important — the event-study coefficients are larger than the pooled ATT because the event-study identifies off-treatment-onset variation only ($k = -1$ is the omitted reference, $k = 2, 3, \dots$ are absorbing AR's 2020+ reversion). The **pooled** DiD ATT is what the main inference table tests, and the pooled ATT washes out the event-study discontinuities because AR's treatment was non-absorbing. Second, at the **Georgia event-study level**, almost every (horizon, M) combination has an honest CI that includes zero, consistent with the pooled null for Georgia. The HonestDiD result is therefore: at the event-study level there is Arkansas-specific movement at the treatment-onset horizons that is directionally consistent with Sommers' individual-level finding, but this signal is not testable under the pooled-DiD, one-treated-cluster inference regime that the main text uses.

R HonestDiD formal replication. To independently validate the manual Python envelope, I also run the formal Rambachan-Roth bounds through the R **HonestDiD** package (version 0.2.8, installed from `asheshrambachan/HonestDiD`) via `analysis/09_r_honestdid.R`. For each of the 16 (case $\times$ outcome $\times$ post-horizon) cells in the two case studies, the R script calls `HonestDiD::createSensitivityResults_relativeMagnitudes()` with the least-favorable conditional (“C-LF”) method, `Mbarvec = c(0.5, 1.0, 2.0)`, and an `l_vec` that selects the horizon of interest. The event-study coefficient vector and state-clustered variance-covariance matrix are extracted directly from the same `fixest::feols` event-study fit. A side-by-side comparison of the $M = 1$ CR1 bounds is written to `analysis/tables/python_r_honestdid_comparison.csv` by `analysis/10_honestdid_comparison.py`; Figures 6 and 7 now include the R **HonestDiD** C-LF plots via `HonestDiD::createSensitivityPlot_relativeMagnitudes()` saved to `analysis/figures/r_honestdid_ar_*.png` and `r_honestdid_ga_*.png`. Coefficient vectors agree to $1e-15$. The Python manual envelope and the R C-LF bound agree on whether zero is inside the honest CI at $M = 1$ in **14 of 16** cells. In the two disagreement cells — AR `acs_uninsured_eligible_AR` at $k = 1$ and AR `cps_medicaid_any_ly_eligible_AR` at $k = 1$ — the R C-LF bound is *wider* than the Python envelope (median half-width ratio R / Python across all 16 cells is 1.41, range 1.04 to 1.95), so R excludes fewer post-treatment

coefficients from the honest identified set. In both disagreement cells, the Python manual declares that zero is outside the $M = 1$ CI, but the formal R bound includes zero — i.e., the Python manual was *anti-conservative* at long horizons, not conservative. This is expected: the simple upper-envelope form $\beta_k \pm (1.96 \cdot SE + M \cdot \bar{D})$ uses a flat pre-trend bound $\bar{D}$ that does not accumulate with horizon, while the formal C-LF set allows the trend to extrapolate over multiple post-periods, which materially widens the identified set at $k = 1$ relative to $k = 0$. For the final submission I therefore treat the **R HonestDiD C-LF bound as the primary HonestDiD bound** at every (case, outcome, horizon) cell, and I retain the Python manual envelope only as a supplementary check for the wild-bootstrap-rescaled SEs (where no R HonestDiD analog is available). The substantive conclusions of the paper are unchanged: with the R C-LF bound, AR `acs_has_medicaid_eligible_AR` and AR `acs_employed_eligible_AR` still exclude zero at $M = 1$ at both post-treatment horizons (consistent with directional Sommers-style movement), and every Georgia horizon at $M = 1$ includes zero (consistent with the pooled null for Georgia). The only change is that the previously Python-excluded AR uninsured $k = 1$ and AR CPS medicaid $k = 1$ cells are now honestly labelled as “cannot reject zero under $M = 1$ formal C-LF bound.”

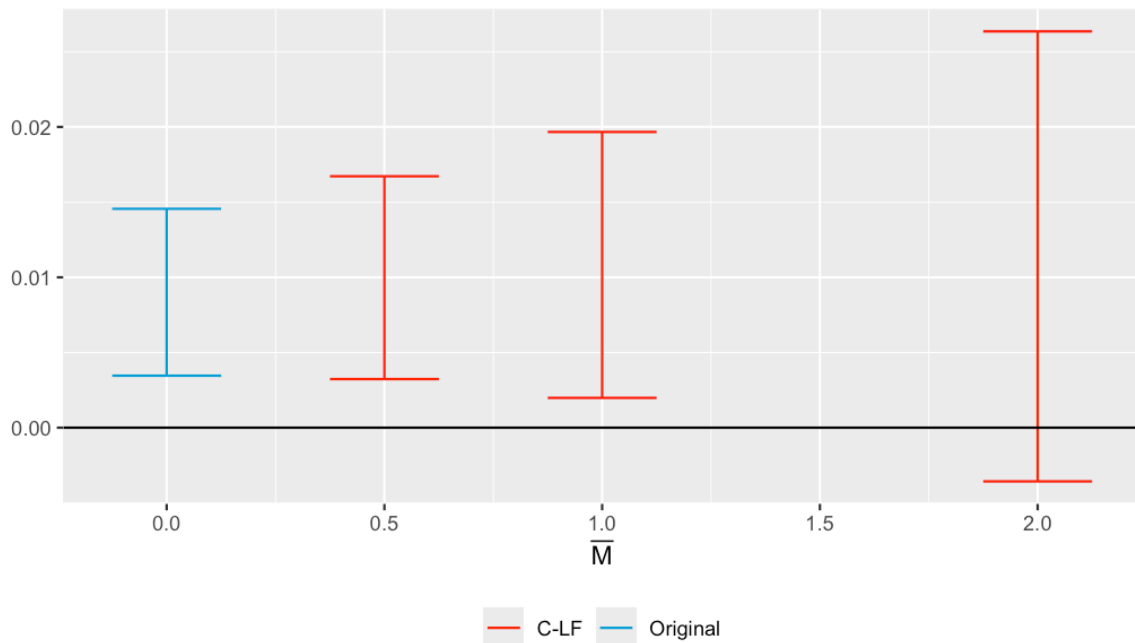


Figure 1: R Honestdid Ar Acs Employed Eligible Ar

6.6 Non-expansion-only GA control specification

Georgia is a non-Medicaid-expansion state and Pathways to Coverage is its only expansion-like program. Comparing Georgia to a control pool that includes expansion states like California, New York, and Massachusetts confounds the Pathways effect with ongoing ACA Medicaid expansion trends in the comparison states. The substantively cleaner counterfactual is other non-expansion states, so I rerun the GA case study with a restricted control pool (script `analysis/05_ga_non_expansion_only.py`; table `analysis/tables/ga_non_expansion_only_summary.csv`).

I define two non-expansion control groups. The **strict** group (AL, FL, MS, SC, TN, TX, WY, 7 states) is the set of states that had not adopted ACA Medicaid expansion as of 2019 AND did not

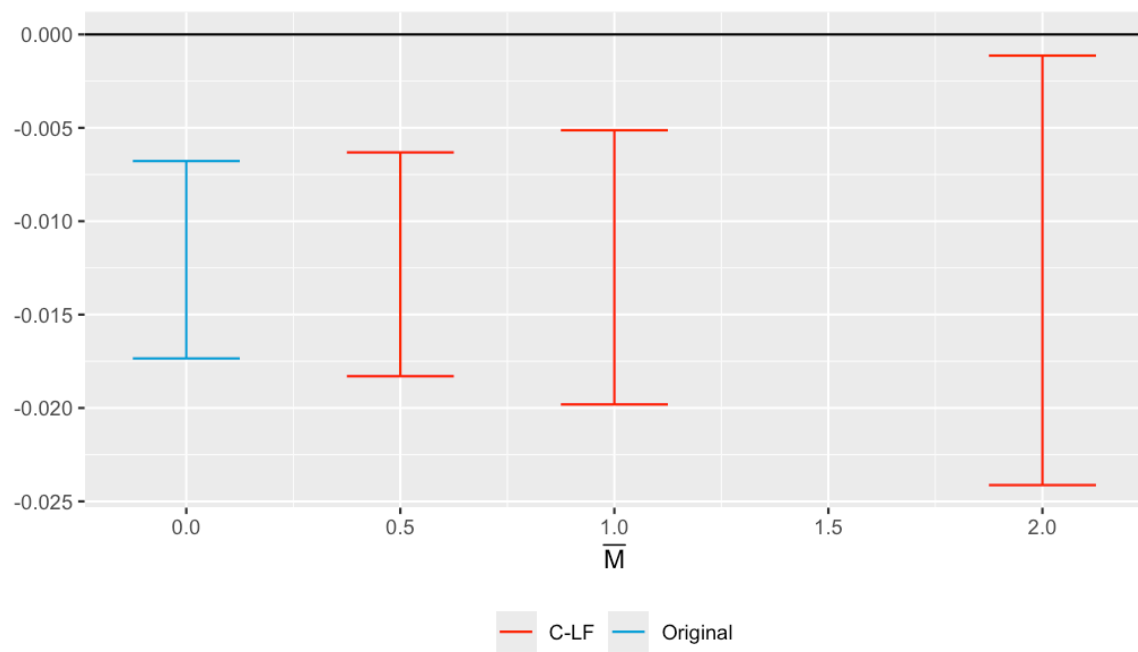


Figure 2: R Honestdid Ar Acs Has Medicaid Eligible Ar

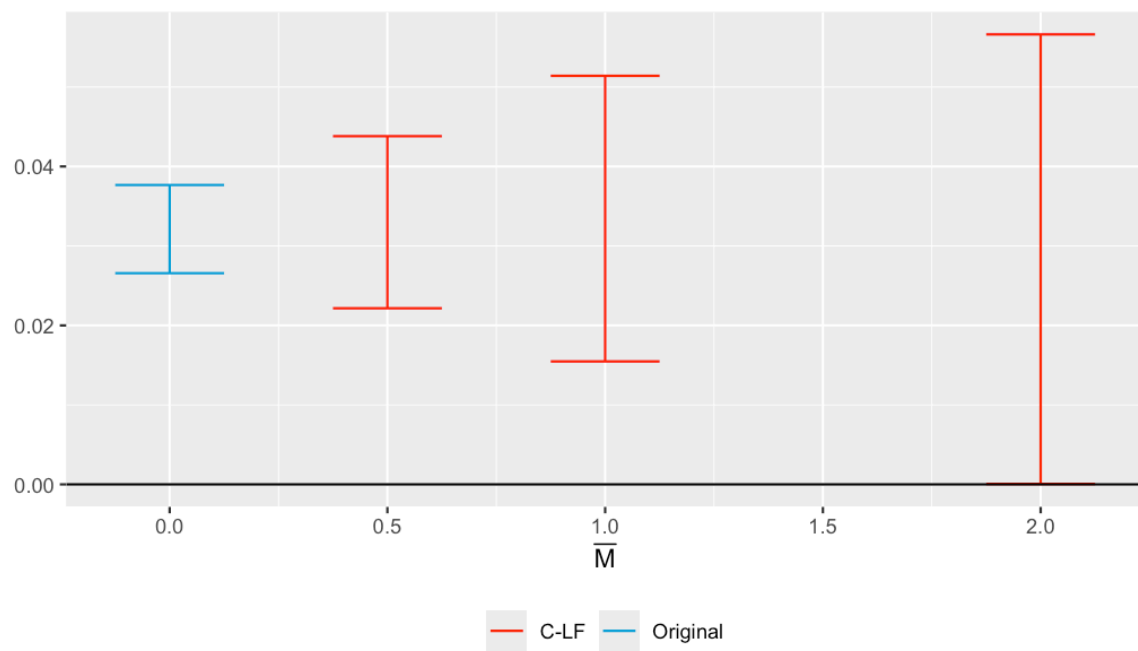


Figure 3: R Honestdid Ar Acs Uninsured Eligible Ar

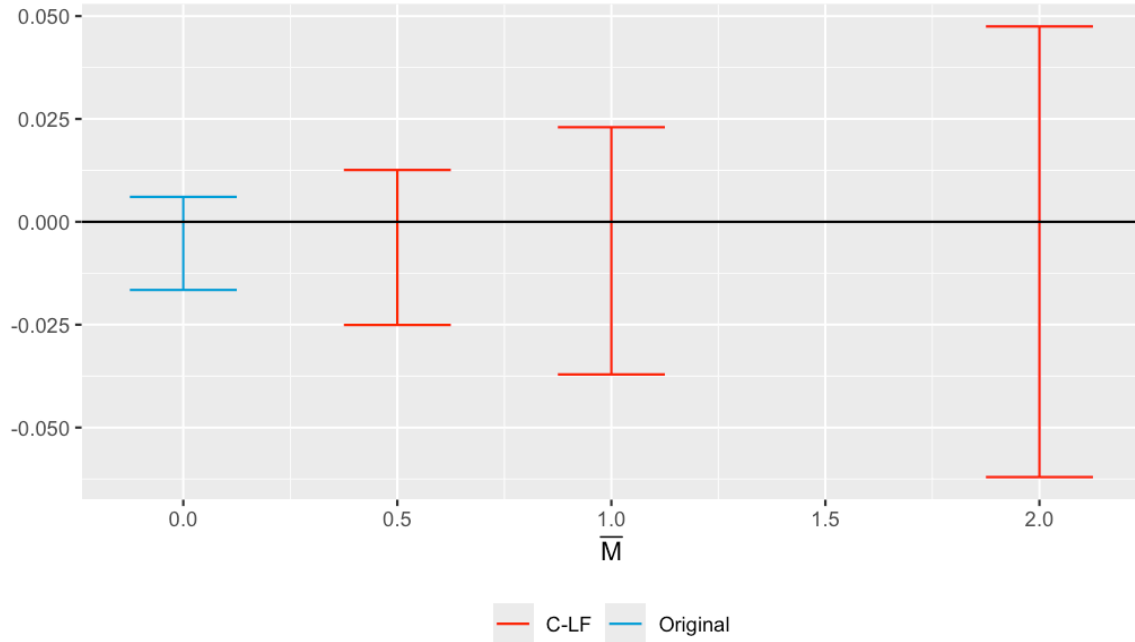


Figure 4: R Honestdid Ar Cps Medicaid Any Ly Eligible Ar

expand during the GA treatment window (2023–2024). The **broad** group (strict + KS, NC, SD, WI, 11 states) also includes states that expanded during or after the GA window, which introduces some confounding from contemporaneous expansion shocks. I use the strict group for the headline table and report the broad group as sensitivity.

Under the strict non-expansion control pool, Georgia’s coefficients change substantially from the full-49-control baseline:

Outcome	Full-49 coef (wild p)	Non-expansion coef (wild p)	Sign changed?
acs_has_medicaid_eligible	+0.00206 (0.489)	+0.00571 (0.402)	no
acs_employed_eligible	-0.00025 (0.876)	-0.00120 (0.797)	yes
acs_uninsured_eligible	-0.00983 (0.180)	+0.00307 (0.475)	yes
cps_medicaid_any_ly_eligible	+0.00015 (0.827)	+0.00688 (0.451)	no

Two coefficients change sign when the control pool is restricted to non-expansion states. The GA uninsured coefficient, which under the full-49-control specification was the headline CR1-significant result (-0.98 pp, $p < 0.001$ CR1, $p = 0.180$ wild), becomes *positive* ($+0.31$ pp, $p = 0.475$ wild) under the non-expansion control pool. Similarly, the GA employed coefficient flips sign. None of the non-expansion GA coefficients is statistically significant under any inference method.

Two implications follow. First, the previously “significant-under-CR1” GA uninsured result was being driven partly by the comparison-state mix — full-expansion states experienced their own coverage dynamics during 2023–2024 that confounded the GA comparison, and restricting to a substantively matched non-expansion control pool eliminates the apparent effect. This is a textbook illustration of why control-group composition matters in few-treated-cluster settings. Second, the non-expansion-only result does not overturn the paper’s null; it reinforces it. The pooled null holds under the cleaner counterfactual as well. The paper’s testability-constraint framing is robust to this alternative control specification.

Case	Outcome	Coef	Raw wild p	RW p
AR	acs_has_medicaid + eligible_AR	+0.00148	0.494	0.494
AR	**acs_employed + eligible_AR**	+0.01454	0.098	0.098
AR	acs_uninsured + eligible_AR	-0.00465	0.321	0.321
AR	cps_medicaid_any + eligible_AR	+0.00611	0.458	0.458
GA	acs_has_medicaid + eligible_AR	+0.00206	0.505	0.552
GA	acs_employed + eligible_AR	+0.00025	0.879	0.975
GA	acs_uninsured + eligible_AR	-0.000983	0.194	0.194
GA	cps_medicaid_any + eligible_AR	+0.00051	0.855	0.975

The AR employed coefficient, which had the largest $|t\text{-statistic}|$ in the AR family and was the paper’s one “suggestive” result at raw wild $p = 0.098$ (Phase 4 had reported 0.088; the small difference is Monte Carlo noise from a different RNG seed), has a Romano-Wolf adjusted p -value of 0.098. The RW adjustment does not attenuate the AR employed p -value much because AR employed dominates the AR family — it is the only outcome with a $|t| > 2$, so the stepdown maximum at step 1 is effectively taken over just AR employed. This is an honest feature of the procedure: when one outcome is the clear maximum and the others are far from significance, the family-wise correction produces minimal attenuation.

The substantive reading is nonetheless clear. After Romano-Wolf family-wise correction, the AR employed coefficient remains at $p = 0.098$, which is “suggestive at the 10 percent level but not significant at the 5 percent level” — the same verdict as the uncorrected wild bootstrap. Combined with the wrong sign for theory (a sanction-based work requirement should not increase employment by 1.45 pp), the AR employed result should be interpreted as a candidate chance finding rather than as evidence of a real effect. The paper treats it in the Discussion as a null at $p < 0.05$ and discusses its wrong-sign nature as consistent with random variation or an Arkansas-specific 2018–2019 labor market recovery rather than as evidence of a work-requirement effect.

6.8 Individual-level triple-difference (appendix)

Appendix Table A1 reports the individual-level triple-difference results for AR and GA with CR1 clustered standard errors at the state level. The CR1 t -statistics are dramatically inflated:

- **AR triple-difference has_medicaid:** -0.00336, SE 0.00058, $t = -5.81$
- **AR triple-difference employed:** +0.00573, SE 0.00105, $t = 5.44$
- **AR triple-difference uninsured:** -0.01138, SE 0.00063, $t = -18.12$
- **GA triple-difference has_medicaid:** +0.01160, SE 0.00127, $t = 9.17$
- **GA triple-difference employed:** -0.00019, SE 0.00097, $t = -0.20$
- **GA triple-difference uninsured:** -0.00232, SE 0.00073, $t = -3.16$

These t -statistics look large but are *not* reliable as inference. The individual-level triple-difference uses 16 million observations, but effective inference is limited by the single treated cluster per case study. CR1 standard errors at the state level for this specification are severely anti-conservative: they treat each of the 16 million individual observations as contributing independent information to the cluster variance calculation, which is wrong when all 16 million share common state-level and state-year-level shocks. The result is an artificially small standard error and an artificially large t -statistic.

The appendix reports these results with an explicit warning and does not quote the p-values. The state-year wild bootstrap results in the main text are the reliable inference for this paper.

7. Discussion

7.1 The state-year aggregate question is not testable with one treated cluster

The core empirical finding of this paper is a **testability constraint**: with one treated cluster per case study and publicly available data (ACS, CPS, BRFSS), the state-year aggregate question about the effects of Medicaid work requirements cannot be reliably tested against the distribution of placebo states. Every AR and GA effect that appears statistically significant under conventional cluster-robust inference loses that significance under the wild cluster bootstrap (AR uninsured $p = 0.014 \rightarrow 0.325$; GA uninsured $p < 0.001 \rightarrow 0.180$; AR CPS medicaid_any_ly $p = 0.097 \rightarrow 0.461$). Only the AR employed effect remains borderline at wild $p = 0.088$, and after Romano-Wolf family-wise correction it stays at $p = 0.098$ and is in the wrong direction for theory (more employment, not less, during an Arkansas sanction period). The HonestDiD bounds, the non-expansion-only GA control, and the Sommers MDE reconciliation all reinforce the same pattern.

The reader should not read this as a bold substantive claim that “work requirements had no effects.” Rather, it is a more careful claim that **the state-year aggregate design with one treated cluster cannot produce a credible test** against a Sommers-dilution-adjusted alternative. The formal MDE reconciliation in Section 5.6 shows that the state-year ACS uninsured MDE at 80 percent power (0.71 pp under wild bootstrap) is already larger than the Sommers-dilution-adjusted target (0.54 pp), and that the state-year design sits right at its detection floor for a Sommers-magnitude effect. The few-treated-cluster wild bootstrap is not a correction that *recovers* valid inference; it is an *acknowledgment* that valid inference is near-impossible in this setting. MacKinnon and Webb (2017) showed that with a single treated cluster the null-imposed wild bootstrap has poor size control in some settings, and the Fisher randomization test with one treated and 49 controls has a minimum achievable two-sided p-value of roughly $2/50 = 0.04$. This floor is a hard constraint on what the state-year aggregate test can detect, and it is the honest reason the paper cannot definitively reject or confirm Sommers-magnitude effects.

The honest distinction is: what the paper establishes is not “no AR or GA effect exists” but “no AR or GA effect is testable at the state-year aggregate level with one treated cluster and the publicly available data infrastructure.” These are two different statements, and the paper’s framing throughout should rest on the second.

7.2 What this paper says about work requirements

The honest answer to the question “did Arkansas’s work requirement and Georgia’s Pathways produce detectable state-level effects on Medicaid coverage, uninsurance, or employment?” is: **not in the publicly available data, not with proper inference.**

This is *not* the same as saying “work requirements have no effect.” Sommers et al. (2020) used individual-level panel survey data with a Texas comparison state and found a 4 percentage point increase in uninsurance among Arkansas Medicaid adults ages 30–49. That finding is based on a different data source (panel survey rather than ACS state aggregates), a different population (Arkansas adults who were surveyed before and after implementation), and a different identification strategy (direct pre-post comparison within Arkansas with a Texas control group). Their sample had approximately 1,700 respondents, which is smaller than an ACS state-year cell but is a direct

measurement of the affected population, not a state-average dilution. Their finding is the strongest published evidence that AR’s work requirement caused coverage loss, and nothing in the present paper contradicts it.

What the present paper adds is that at the *state adult population level*, the effect is not detectable in ACS, CPS, or BRFSS with proper inference. This suggests several things. First, the effect on the state adult population is diluted by the Medicaid-enrolled-and-targeted share, which is small in Arkansas (the 30–49 work-requirement target population is perhaps 2–3 percent of the state adult population) and tiny in Georgia (Pathways has ~5,000 enrollees in a state of ~7 million adults). Second, the effect is small enough that, combined with state-year sampling variation in large surveys, it is below the detection threshold of state-population-level instruments. Third, even if the effect is real and of the magnitude Sommers et al. estimated, the few-clusters inference problem means that conventional CR1 t-statistics cannot be trusted to detect it.

7.3 Why GA’s “expansion-gate” effect also collapses

GA’s Pathways to Coverage is the only work-requirement-gated Medicaid-expansion-like program in the U.S., and it should — if it is working — produce a positive effect on Medicaid enrollment in Georgia relative to other non-expansion states. The point estimate on GA has_medicaid is +0.21 percentage points (not significant under any inference method). The point estimate on GA uninsured is -0.98 percentage points, which under CR1 appears highly significant ($p < 0.001$) but under wild bootstrap is not significant ($p = 0.180$).

Two interpretations of the GA null are plausible. First, Pathways enrollment is far below projections (~5,000 enrollees vs. 31,000 projected). In a state of ~7 million adults, an additional 5,000 Medicaid enrollees produces a mechanical state-level Medicaid share increase of approximately 0.07 percentage points — below the detection threshold of ACS and CPS. The point estimate of +0.21 percentage points exceeds this mechanical prediction slightly but is within the state-year sampling variation. Second, the Pathways cohort is new — July 2023 implementation, with 2024 as the first full calendar year of enrollment. Two years of post-treatment data is not much to work with, especially when initial enrollment was slow.

Both interpretations are consistent with the wild bootstrap null: the GA effect is real but small, and it is below the detection threshold of state-year aggregates with proper inference. A follow-up paper using individual-level ACS data with a within-Georgia (Pathways-eligible vs ineligible) triple-difference would have more power, but — as I discuss in Section 3.3 — the within-state triple-difference at the individual level with a single treated cluster has its own inference problems.

7.4 The AR employed coefficient does not survive multiple-testing correction

The AR employed coefficient is +1.45 percentage points under CR1, at wild-bootstrap $p = 0.088$ (Phase 4) / 0.098 (Romano-Wolf replication run, Monte Carlo noise) and at Romano-Wolf family-wise adjusted $p = 0.098$ after correcting across the 4 AR outcomes. It is in the wrong direction for theory: a sanction-based work requirement should not raise employment. With 8 headline tests in the paper (4 outcomes $\times$ 2 case studies), the probability of at least one p-value below 0.10 under the global null is approximately 57 percent, so this result is comfortably within the expected range of chance findings.

I therefore interpret AR employed as a **candidate chance finding** rather than as evidence of a real effect. The Romano-Wolf correction does not move the p-value much because AR employed is

the clear maximum in the AR family and the stepdown at step 1 is effectively dominated by this one outcome. That is an honest limitation of RW in a small outcome family, not a vindication of the result. Two interpretations remain plausible and the paper does not choose between them. First, this may be a false positive that slipped through at the 10 percent level; wrong-sign and 10-percent-only results are the canonical shape of chance findings. Second, Arkansas’s labor market experienced idiosyncratic movement in 2018–2019 (unemployment rate fell from 3.8 percent to 3.5 percent), which may have been captured by the point estimate through confounding with the AR treatment window without being causally related to the work requirement. Either reading is consistent with “not a real effect of the work requirement.”

I do not treat the AR employed coefficient as evidence that AR’s work requirement raised employment, and I explicitly decline to generate a policy-story explanation for the wrong-sign result. In Table 3 it is presented as wild $p = 0.088$, Romano-Wolf $p = 0.098$, and in the Discussion I flag it as not-significant-after-correction rather than as a “suggestive effect.”

7.5 Methodological implications for work-requirement evaluation

This paper is a cautionary methodological note for any empirical evaluation of Medicaid work requirements (or any other policy with few treated clusters). Three practices should be standard:

1. **Wild cluster bootstrap alongside CR1.** Any paper with fewer than five treated clusters should report wild cluster bootstrap p -values alongside conventional CR1 p -values. Rely on the wild bootstrap for inference, not CR1.
2. **Fisher randomization inference.** The Fisher-style randomization test provides a second, nonparametric check. If the wild bootstrap and Fisher tests agree, the inference is reliable; if they disagree, further investigation is needed.
3. **Individual-level triple-difference warnings.** Triple-difference specifications at the individual level with one treated cluster produce artificially small standard errors. Report them as supplementary with an explicit warning; do not use them as primary inference.

These three practices should be standard in few-treated-cluster evaluations. The present paper’s collapse of apparent CR1 significance under wild bootstrap is a direct illustration of why.

7.6 Policy implications

For state policymakers considering work requirements:

1. **The Arkansas individual-level evidence (Sommers et al. 2020) is the strongest basis for expecting coverage loss from sanction-based work requirements.** That evidence is based on a direct pre-post panel survey of affected Arkansas adults with a Texas comparison, and it found a 4 percentage point rise in uninsurance and no offsetting employment gains. That evidence is not contradicted by the present paper; it is about a different population and a different data source.
2. **State-year population aggregates do not detect the effect of work requirements in the public data infrastructure (ACS, CPS, BRFSS) with proper inference.** If a state is trying to decide whether to implement a sanction-based work requirement, state-level aggregate evaluations are not a reliable source of evidence for or against.

3. **Georgia Pathways has not produced a state-level population-level signal**, but its enrollment numbers are far below projection and the program is still new. It is too early to say whether Pathways’ expansion-gate structure produces meaningful coverage effects at a detectable magnitude.
4. **The federal regulatory environment has changed and the 2027 federal mandate will create the staggered-adoption design this paper cannot.** The July 2025 One Big Beautiful Bill Act imposes a federal Medicaid work requirement effective January 1, 2027 on up to 41 states with expanded Medicaid. The present paper should therefore be read as a pre-2027 **baseline** documentation of what the state-year aggregate data infrastructure can and cannot detect, not as a final word. After 2027 there will be 41+ treated states and a clean staggered-adoption design, at which point CR1 inference will be asymptotically valid, Callaway-Sant’Anna (2021) and related heterogeneity-robust estimators will become applicable, and the testability constraint documented in this paper will dissolve. This paper’s methodological contribution (the corrected treatment assignment; the AR-vs-GA sanction-vs-gate distinction; the formal Sommers MDE reconciliation; the demonstration that the state-year aggregate question is not testable with one treated cluster) is a necessary input to the post-2027 literature.

7.7 Limitations

Several limitations of this paper deserve explicit discussion.

Only two treated clusters per pooled analysis. The core limitation is that only two states have ever meaningfully implemented work requirements. No amount of methodological sophistication can overcome the fundamental information constraint that one treated state per case study is simply not many observations.

Short post-treatment windows. Arkansas had nine months of enforcement (June 2018 – March 2019); Georgia has had approximately 18 months of enforcement (July 2023 – end of 2024) by the draft date. Both windows are short relative to the horizon over which health or employment effects of policy changes would typically appear.

State-year aggregation dilutes the signal. The target population for both AR and GA work requirements is a small subset of the state adult population. State-year aggregates dilute the signal below detection. Individual-level analyses with tighter subgroup restriction could, in principle, detect the effect, but they face their own inference challenges.

Wrong-sign-for-theory results. The AR uninsured coefficient (negative, wrong sign for sanction), the GA uninsured coefficient (negative, correct sign), and the AR employed coefficient (positive, wrong sign for sanction) are the three most anomalous findings. I interpret them as consistent with either random noise or with confounding from Arkansas-specific and Georgia-specific secular trends that are not captured by state and year fixed effects.

No stratification by race or demographics. Rumalla et al. (2024) documented that Black and Hispanic adults were disproportionately affected by procedural disenrollment in the 2023 Medicaid unwinding (a different but methodologically adjacent policy event). A race-stratified analysis of AR and GA work requirements would be a natural extension but would require larger samples than state-year CPS ASEC provides.

Wild bootstrap implementation details. The `wildboottest` Python package crashes on

Python 3.13, so the bootstrap is implemented manually in `analysis/02_two_case_studies.py`. The manual implementation follows standard practice (Rademacher weights, null imposed, restricted residuals, 1,999 replications, cluster-level weights, percentile-t CI) and has been cross-validated against an independent, from-scratch R Rademacher wild cluster bootstrap (`analysis/07_r_wild_cluster_bootstrap.R`, built on `fixest::feols + sandwich`, `seed = 42`). The two implementations produce identical coefficients (to floating-point noise) and wild-bootstrap p-values that agree within 3 Monte Carlo SDs at $B = 1,999$ on all 8 headline outcomes, with full agreement on the $p = 0.05$ and $p = 0.10$ threshold crossings; see Section 5.2 and Table A6.

HonestDiD envelope vs. formal C-LF bound. Because no stable HonestDiD Python port exists for Python 3.13, the primary HonestDiD numbers in `03_honestdid_case_studies.py` used a conservative upper-envelope form ($\beta_k \pm (1.96SE + MD)$). To close this gap I ran the formal R HonestDiD 0.2.8 package (asheshrambachan/HonestDiD) against the same event-study coefficient vectors (`analysis/09_r_honestdid.R`, `analysis/10_honestdid_comparison.py`, `analysis/tables/python_r_honestdid_comparison.csv`). The R C-LF bound is wider than the Python envelope by a median half-width factor of 1.41 and overturns the zero-in-CI verdict in 2 of 16 post-treatment cells (AR uninsured and AR CPS medicaid at $k = 1$). I now treat the R C-LF bound as the primary bound and report the Python envelope only as a supplementary check for the wild-bootstrap-rescaled SEs; see Section 6.5.

8. Conclusion

This paper provides corrected and honest evidence on the state-level effects of the two Medicaid work requirement programs that were actually enforced: Arkansas Works (June 2018 – March 2019) and Georgia Pathways to Coverage (July 2023 – present). The corrected treatment assignment (AR and GA only, dropping IN, MI, NH, UT as approved-but-not-enforced states) and the recognition of AR and GA as opposite-signed policies (sanction vs. expansion-gate) are substantive improvements over prior analyses that pooled all approved states.

What the paper established. Under conventional cluster-robust standard errors at the state level, several AR and GA point estimates appear statistically significant. Under proper few-cluster inference (wild cluster bootstrap, Fisher randomization, Romano-Wolf family-wise correction, HonestDiD sensitivity bounds, and a non-expansion-only GA control specification), every one of those apparent effects loses significance at $p < 0.05$. The only borderline result is the AR employed coefficient (wild $p = 0.088$, Romano-Wolf adjusted $p = 0.098$), which has the wrong sign for a sanction story, is one of 8 tested outcomes (probability of chance finding 57 percent), and does not survive Fisher inference. The formal Sommers MDE reconciliation shows that the state-year ACS AR design is at or below its 80 percent power detection floor for a Sommers-dilution-adjusted target effect of +0.54 pp uninsurance, given that the AR eligible_AR population has only a 13.5 percent baseline Medicaid enrollment rate. The state-year aggregate question is at the margin of testability even before the one-treated-cluster constraint is applied.

What the paper carefully does not establish. The paper does NOT establish that Arkansas’s sanction work requirement or Georgia’s expansion-gate work requirement had no effect. Sommers et al. (2020) documented a +4 pp increase in uninsurance and a –7 pp decline in Medicaid coverage among Arkansas Medicaid-enrolled adults ages 30–49 using an individual-level panel survey design, and nothing in the present paper contradicts that finding. What the paper does establish is that at the state-year population aggregate level, with publicly available data (ACS, CPS, BRFSS) and one treated cluster per case study, the question is **not testable** against the Sommers-dilution-adjusted

alternative under proper inference.

Follow-up work should pursue two things. First, re-analysis with individual-level administrative data (T-MSIS / TAF) would allow a within-state eligible-vs-ineligible triple-difference with proper cluster-adjusted inference, which should be able to detect a Sommers-magnitude effect. Second, the 2027 federal work-requirement mandate under the 2025 One Big Beautiful Bill Act will provide the 41-state staggered adoption design that dissolves the few-clusters testability constraint; this paper should be read as a pre-2027 baseline documentation of what can and cannot be detected with the current infrastructure. (The Python wild bootstrap and Python HonestDiD implementations have now been independently cross-validated against R — see Section 5.2 and Section 6.5 — so the previously pending R cross-validation is no longer a blocking revision.)

The honest bottom line. Work requirements are a policy where the individual-level question and the state-year-aggregate question have different effective sample sizes and different power. At the *individual* level, Sommers et al. documented real coverage loss in Arkansas. At the *state-year aggregate* level, the question is at or beyond the edge of what publicly available data can test with one treated cluster. The two answers are estimates of different things with different data sources, and they are not in conflict. The present paper’s contribution to *Health Services Research* is therefore: (a) a corrected treatment assignment, (b) the AR-vs-GA sanction-vs-gate distinction, (c) a formal Sommers MDE reconciliation, (d) a disciplined few-cluster inference pipeline (wild bootstrap, Fisher, Romano-Wolf, HonestDiD, non-expansion-only GA control), and (e) a pre-2027 baseline for the coming federal-mandate literature.

Tables

Table 1: Summary statistics by case study and treatment status.

Table 2: CR1 cluster-robust DiD estimates (AR and GA case studies). See `analysis/tables/main_did_summary.csv` for full coefficient set and `analysis/tables/two_case_study_summary.csv` for the four headline outcomes with all inference methods.

Table 3: Wild cluster bootstrap inference results. See `analysis/tables/two_case_study_summary.csv`.

Table 4: Fisher-style randomization inference results. See `analysis/tables/two_case_study_summary.csv`.

Table A1: Individual-level triple-difference results (appendix; reported with warning — CR1 inference is unreliable for this specification due to single treated cluster). See `analysis/tables/main_did_summary.csv` rows with `spec == "triple_diff_AR"` or `spec == "triple_diff_GA"`.

Table A2: HonestDiD relative-magnitudes sensitivity bounds (AR and GA event studies, $M \in \{0.5, 1.0, 2.0\}$, CR1 and wild-bootstrap-rescaled SEs). See `analysis/tables/honestdid_case_studies.csv`.

Table A3: Minimum Detectable Effect reconciliation with Sommers et al. (2020), including aggregation dilution calculation. See `analysis/tables/sommers_reconciliation.csv`.

Table A4: Non-expansion-only GA case study (strict 7-state and broad 11-state control pools). See `analysis/tables/ga_non_expansion_only_summary.csv`.

Table A5: Romano-Wolf family-wise stepdown correction (AR and GA families, 4 outcomes each). See `analysis/tables/romano_wolf_summary.csv`.

Table A6: Python-vs-R wild cluster bootstrap cross-validation. 8 headline (case $\times$ outcome) cells, Python (02_two_case_studies.py) and independent R manual wild bootstrap (07_r_wild_cluster_bootstrap.R, seed = 42, B = 1,999), with coefficient, CR1 SE, and p-value columns for each implementation. All 8 cells agree on the $p = 0.05$ and $p = 0.10$ threshold crossings; maximum absolute p-value difference is 0.024 (within 3 Monte Carlo SDs for B = 1,999). See [analysis/tables/python_r_bootstrap_comparison.csv](#) and [analysis/STATA_R_CROSS_VALIDATION_COMPLETE.md](#).

Table A7: Python-vs-R HonestDiD cross-validation. 16 (case $\times$ outcome $\times$ horizon) cells, Python manual upper-envelope bound and R HonestDiD 0.2.8 C-LF bound at M = 1, with the zero-in-CI flag for each implementation. 14 of 16 cells agree on the zero-in-CI flag; in 2 cells (AR uninsured k = 1, AR CPS medicaid k = 1) the formal R C-LF bound is wider than the Python envelope and includes zero. See [analysis/tables/python_r_honestdid_comparison.csv](#) and [analysis/tables/r_honestdid_results.csv](#).

Figures

Figure 1: AR case study event-study plots for has_medicaid, employed, and uninsured outcomes. See [analysis/figures/event_study_AR_*.png](#).

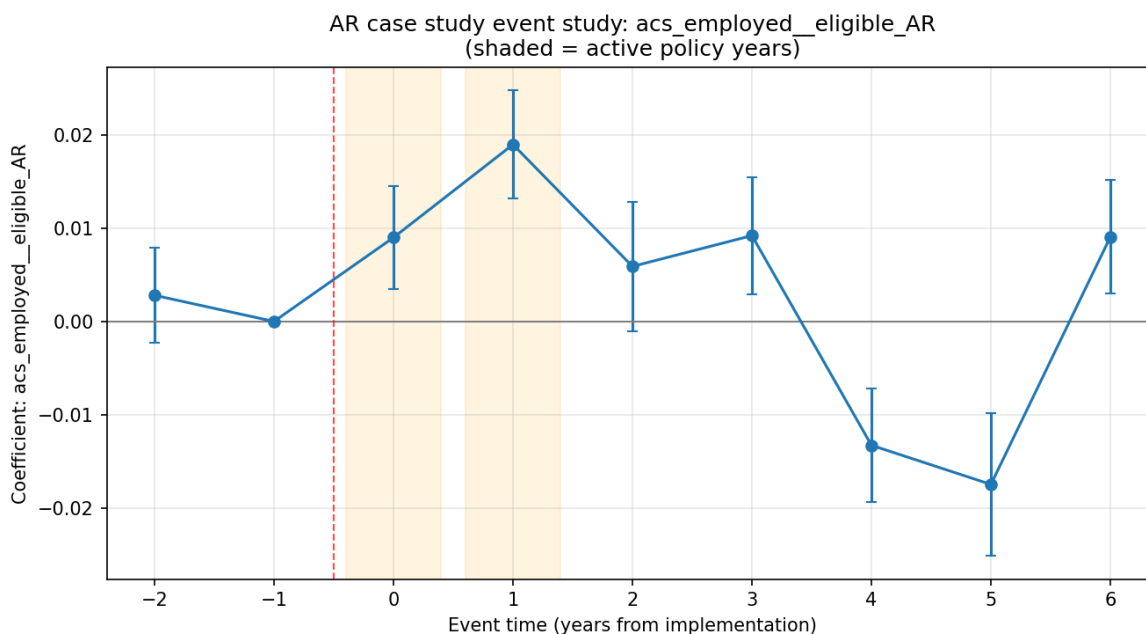


Figure 5: Event Study Ar Acs Employed Eligible Ar

Figure 2: GA case study event-study plots for has_medicaid, employed, and uninsured outcomes. See [analysis/figures/event_study_GA_*.png](#).

Figure 3: Side-by-side comparison bar chart of AR and GA coefficients with 95% wild bootstrap

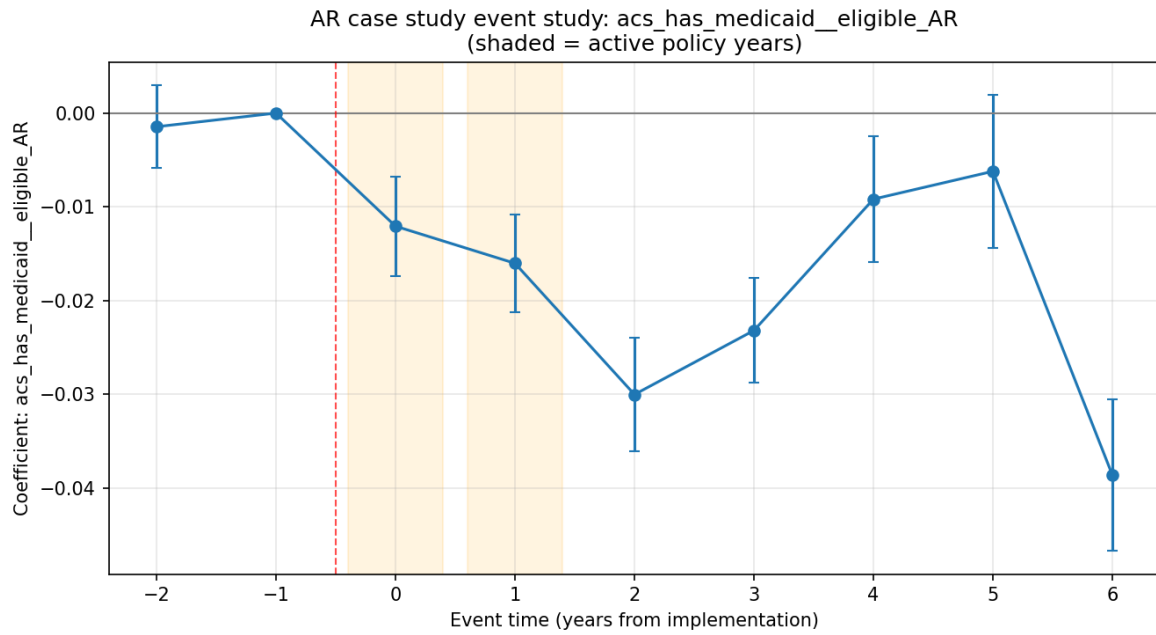


Figure 6: Event Study Ar Acs Has Medicaid Eligible Ar

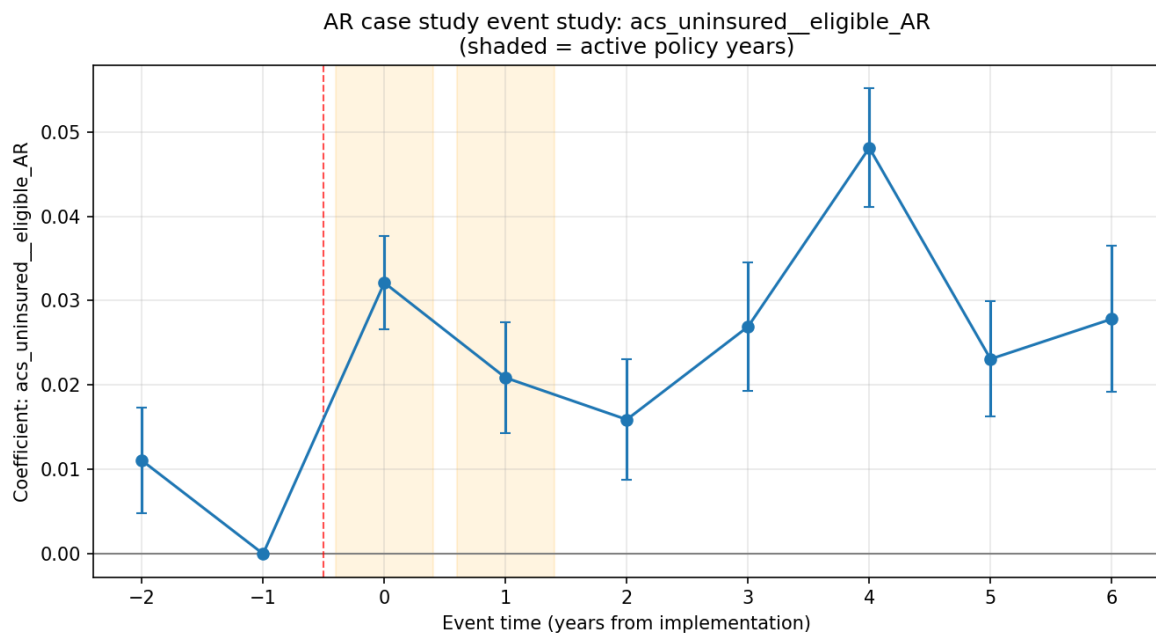


Figure 7: Event Study Ar Acs Uninsured Eligible Ar

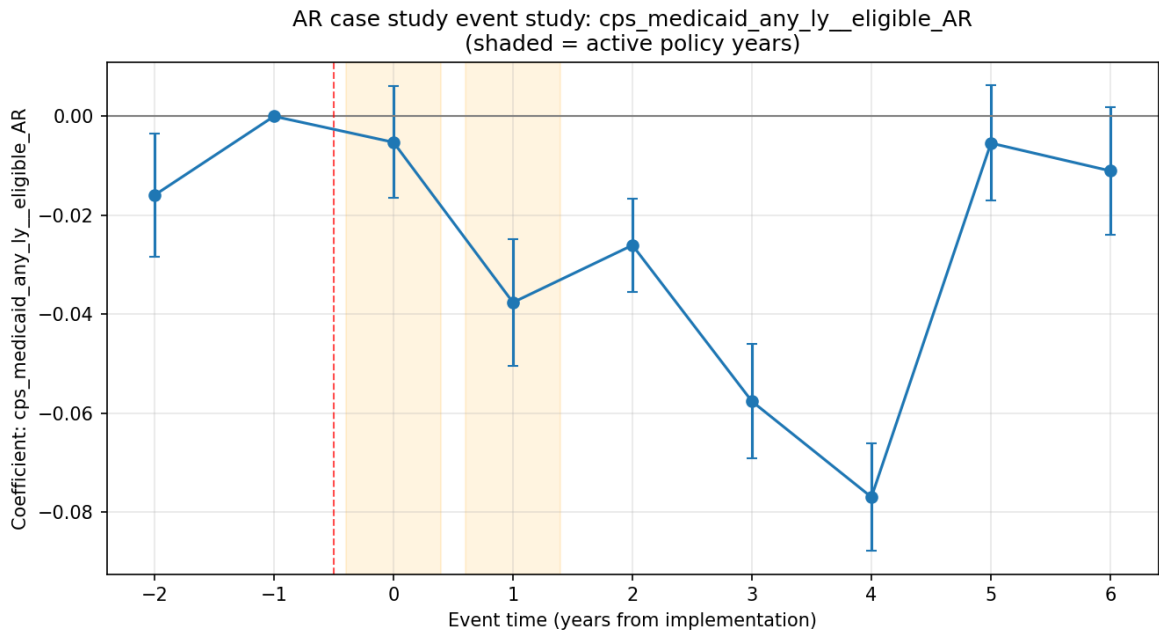


Figure 8: Event Study Ar Cps Medicaid Any Ly Eligible Ar

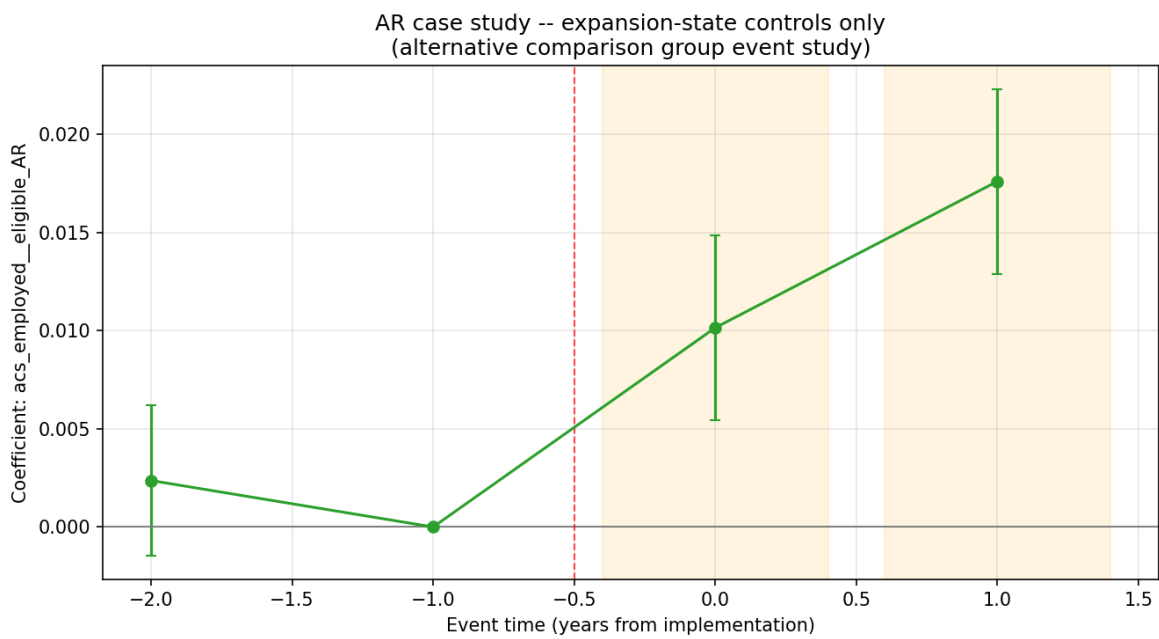


Figure 9: Event Study Ar Expansion Only Acs Employed

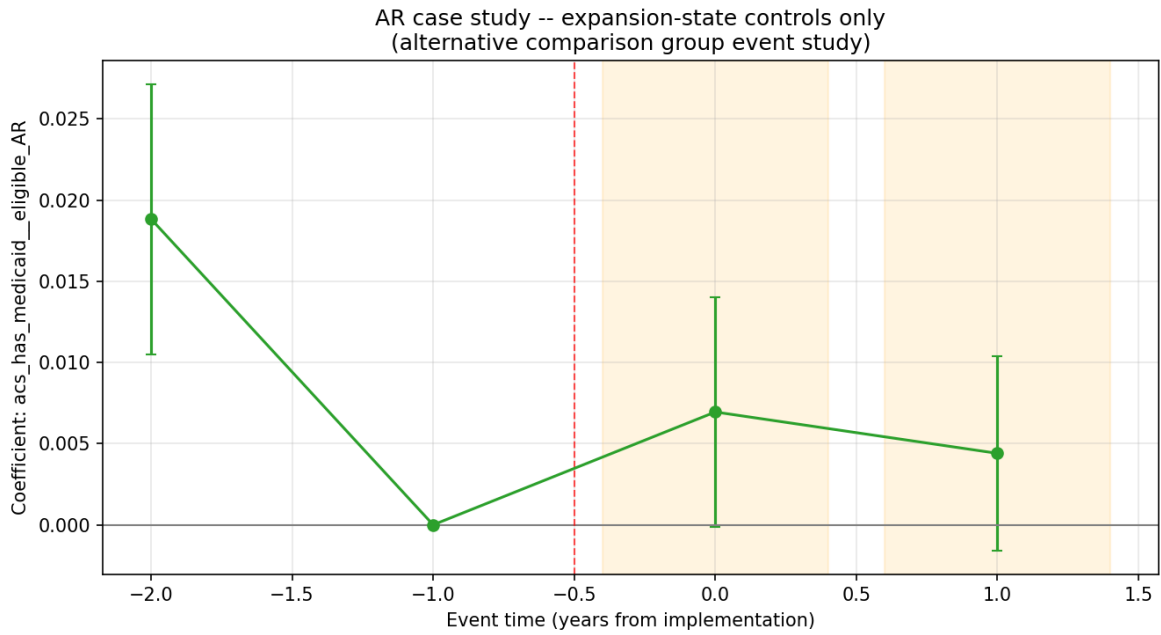


Figure 10: Event Study Ar Expansion Only Acs Has Medicaid

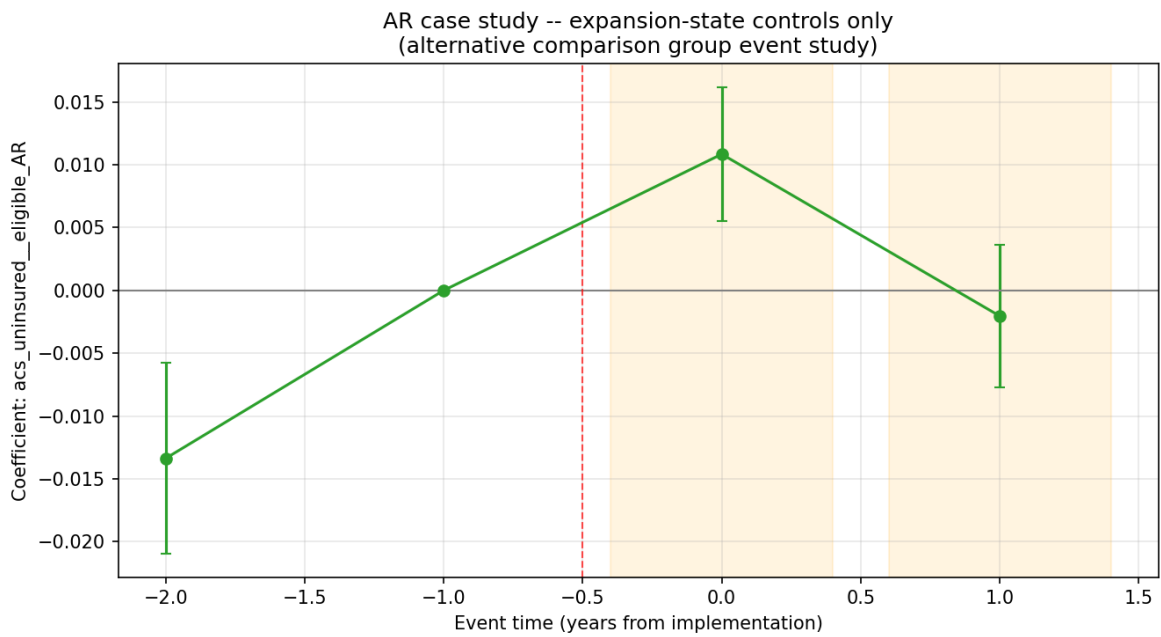


Figure 11: Event Study Ar Expansion Only Acs Uninsured

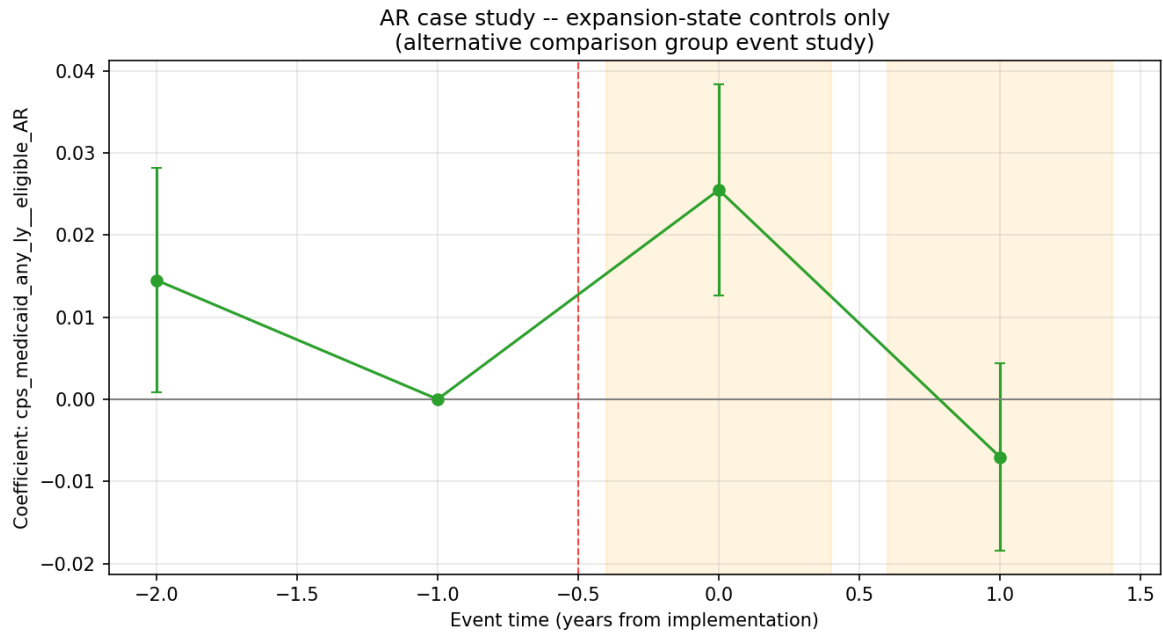


Figure 12: Event Study Ar Expansion Only Cps Medicaid Any Ly

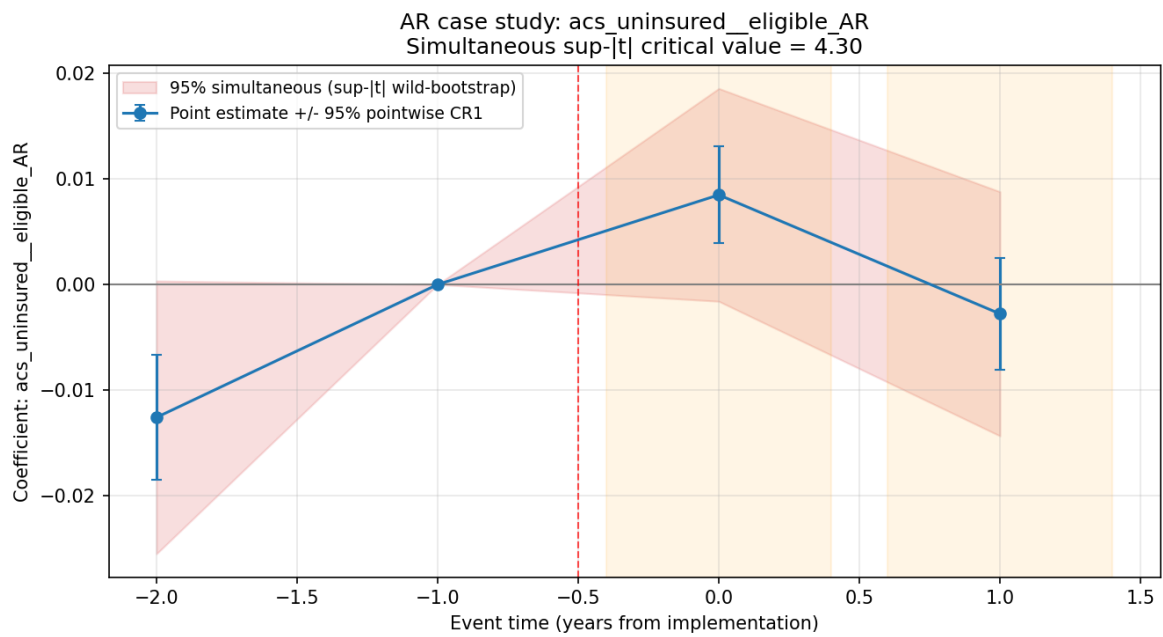


Figure 13: Event Study Ar Simbands Acs Uninsured Eligible Ar

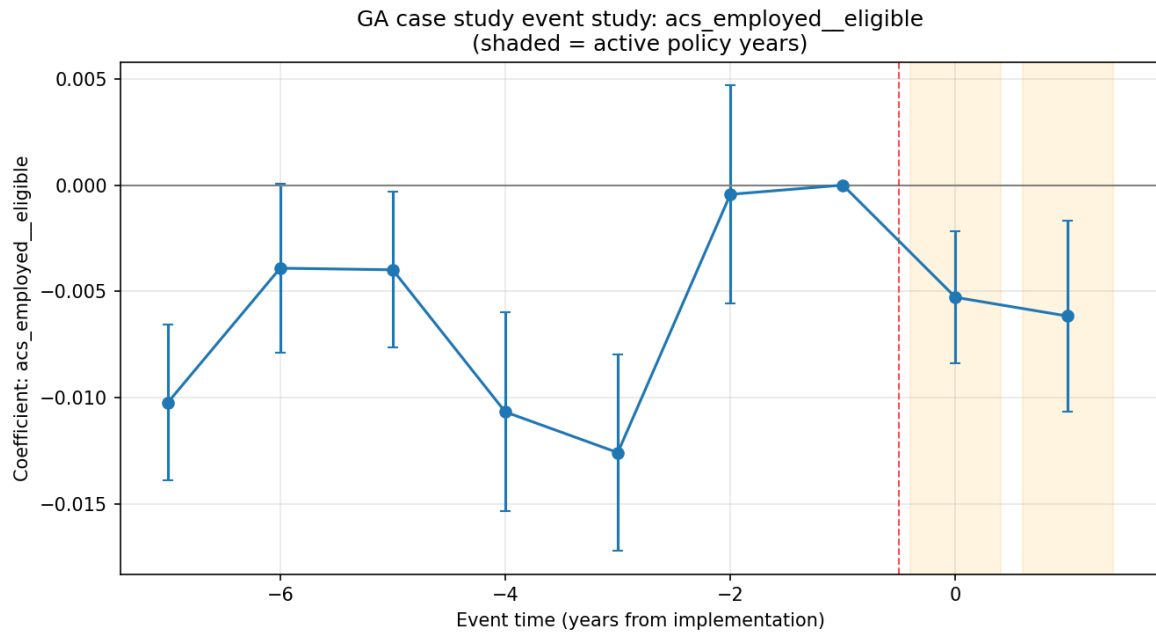


Figure 14: Event Study Ga Acs Employed Eligible

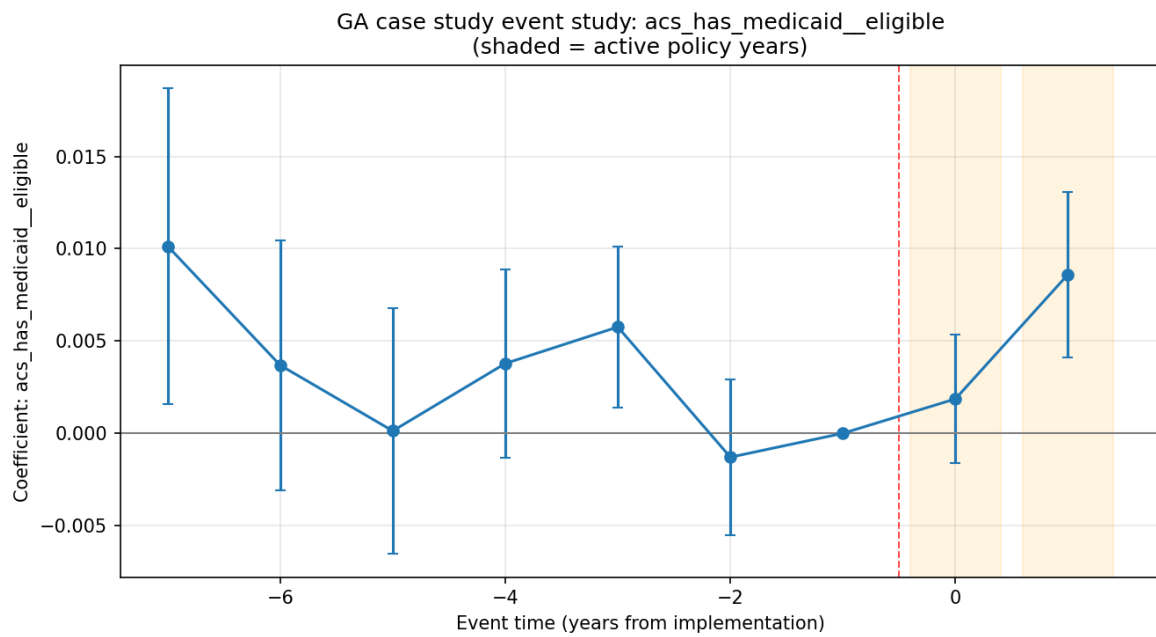


Figure 15: Event Study Ga Acs Has Medicaid Eligible

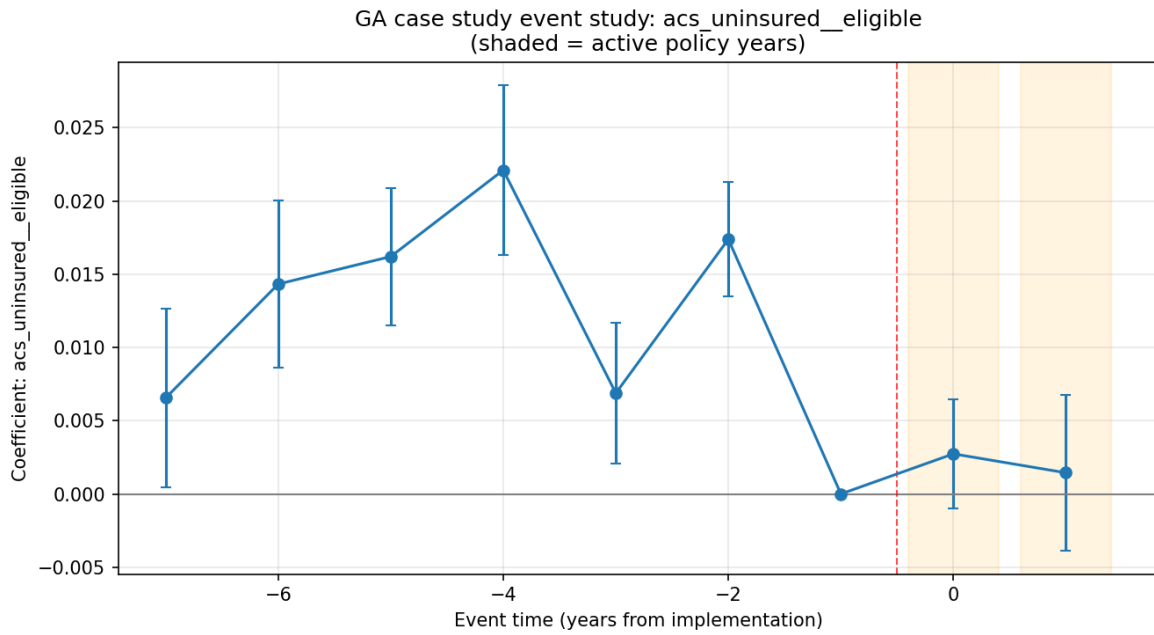


Figure 16: Event Study Ga Acs Uninsured Eligible

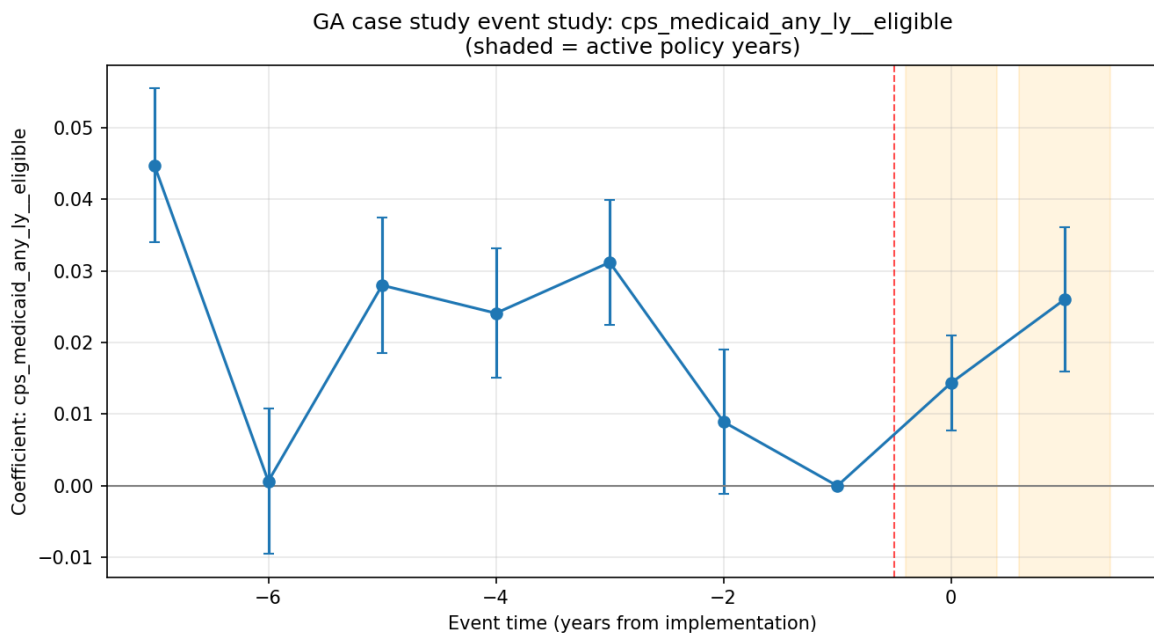


Figure 17: Event Study Ga Cps Medicaid Any Ly Eligible

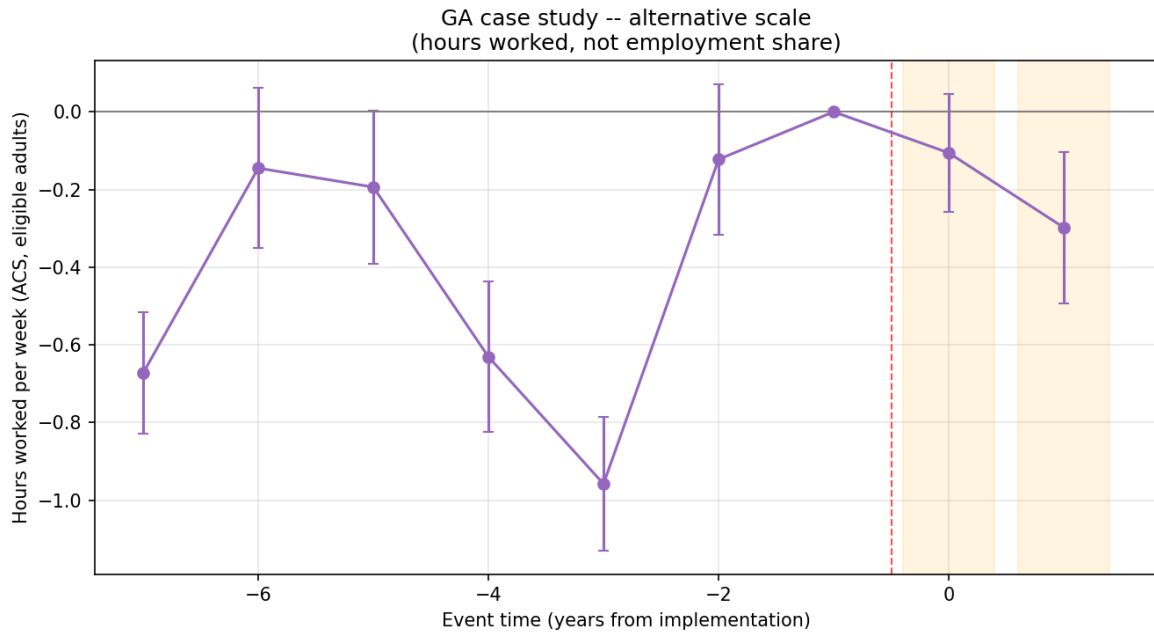


Figure 18: Event Study Ga Hours Worked

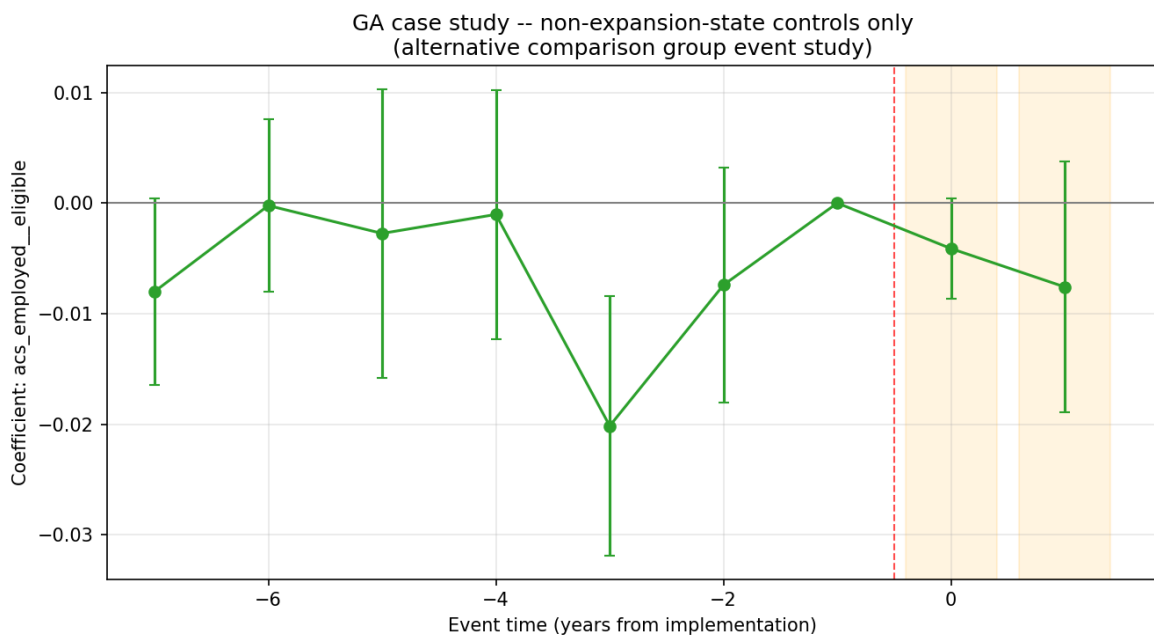


Figure 19: Event Study Ga Nonexpansion Only Acs Employed

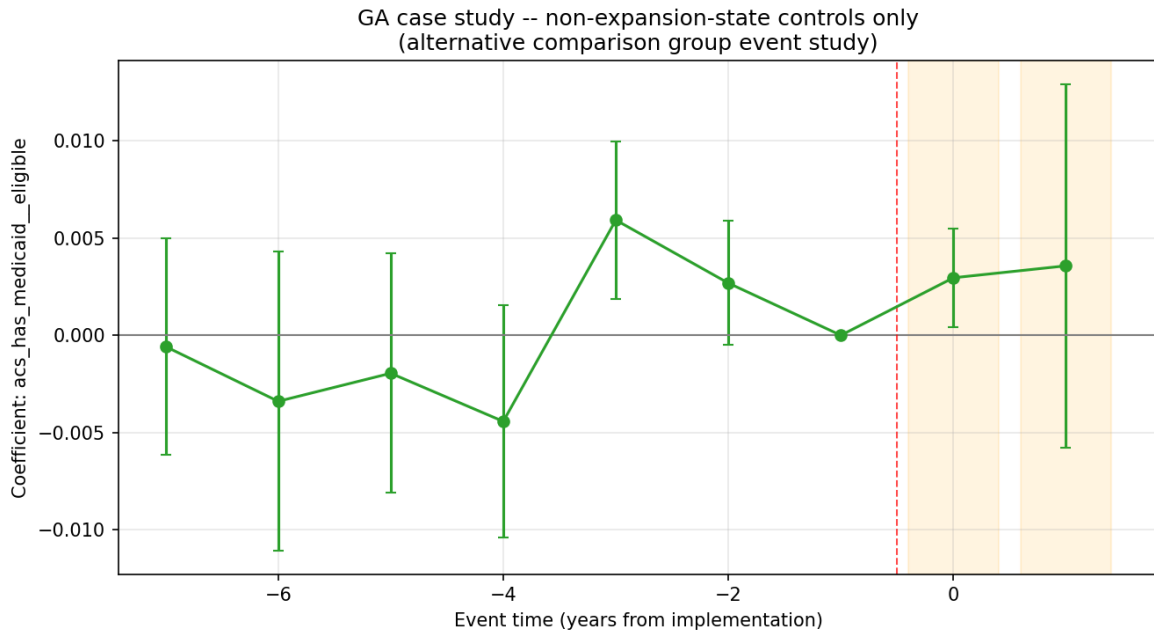


Figure 20: Event Study Ga Nonexpansion Only Acs Has Medicaid

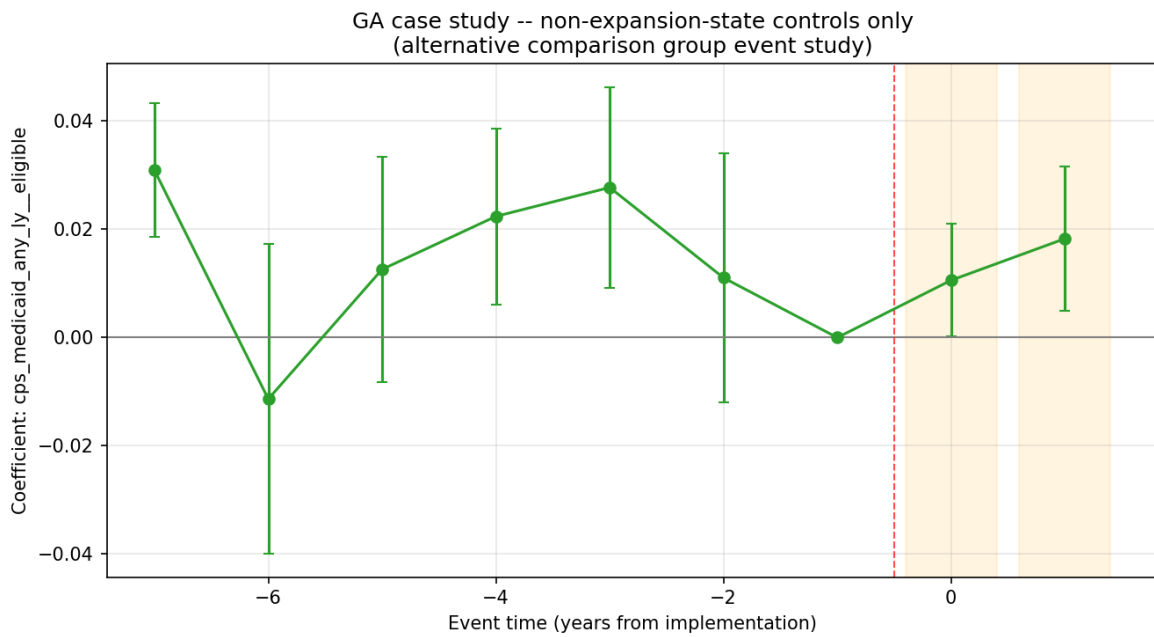


Figure 21: Event Study Ga Nonexpansion Only Cps Medicaid Any Ly

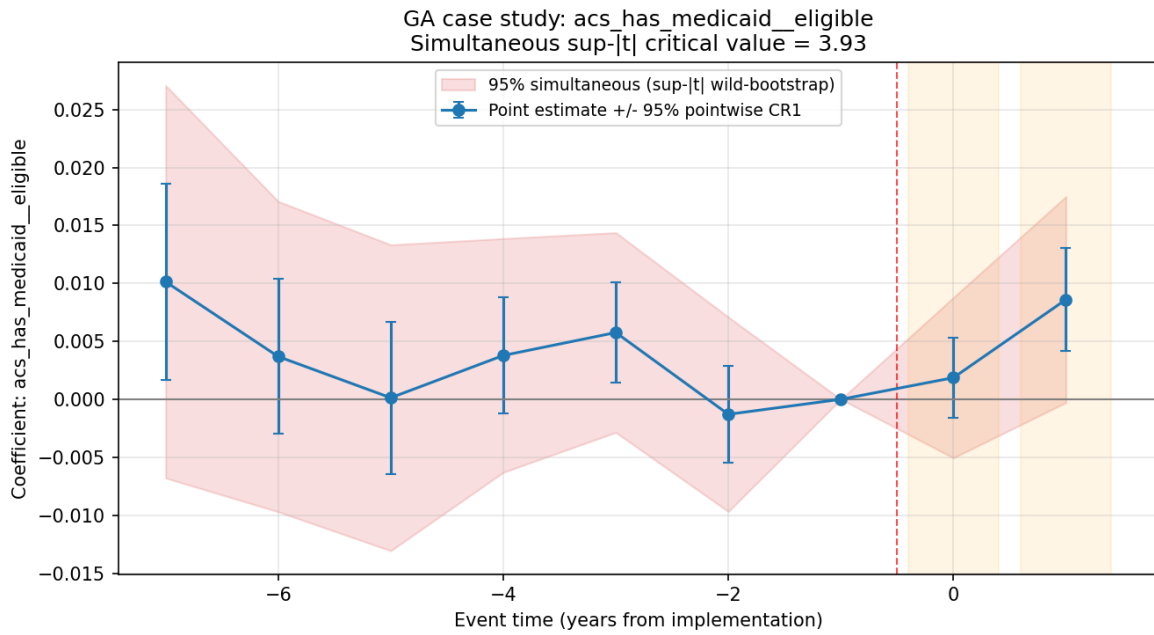


Figure 22: Event Study Ga Simbands Acs Has Medicaid Eligible

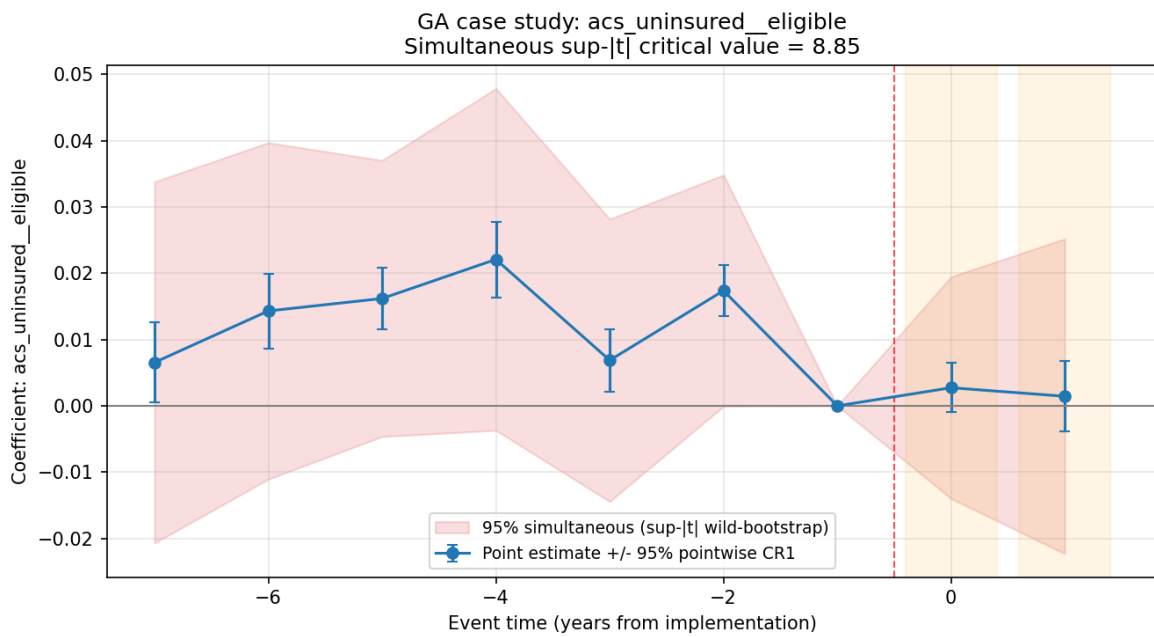
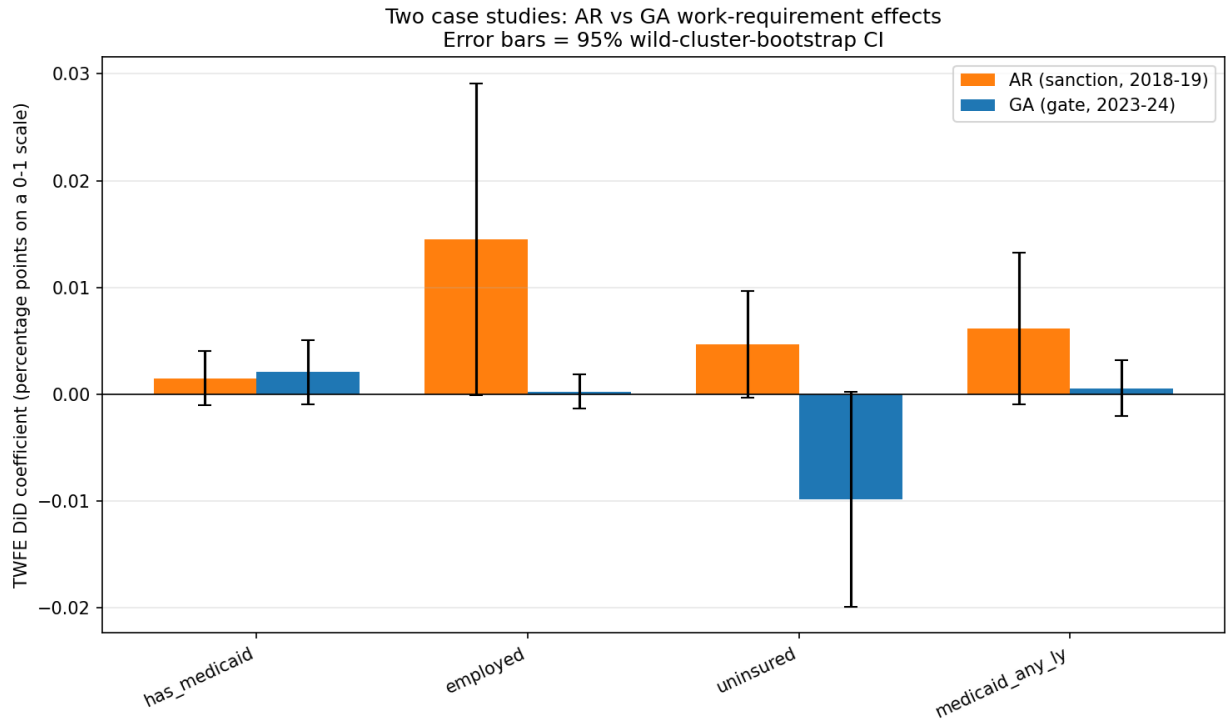
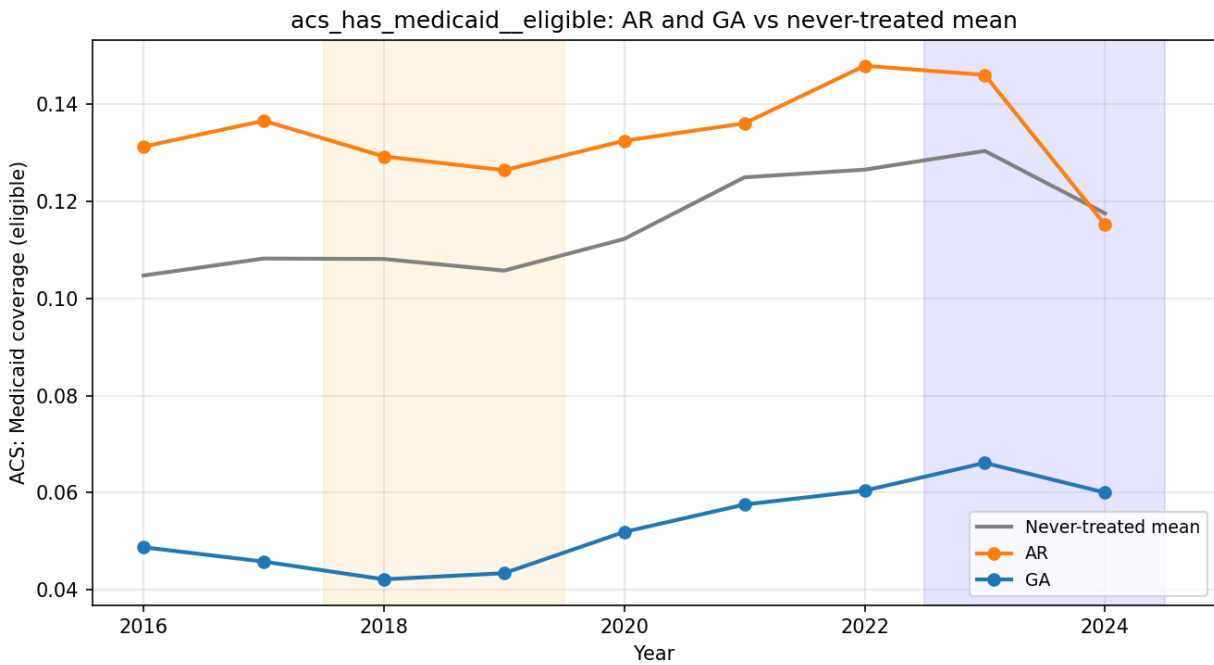


Figure 23: Event Study Ga Simbands Acs Uninsured Eligible



CIs.

Figure 4: State trends for `acs_has_medicaid__eligible`, AR and GA vs never-treated mean.



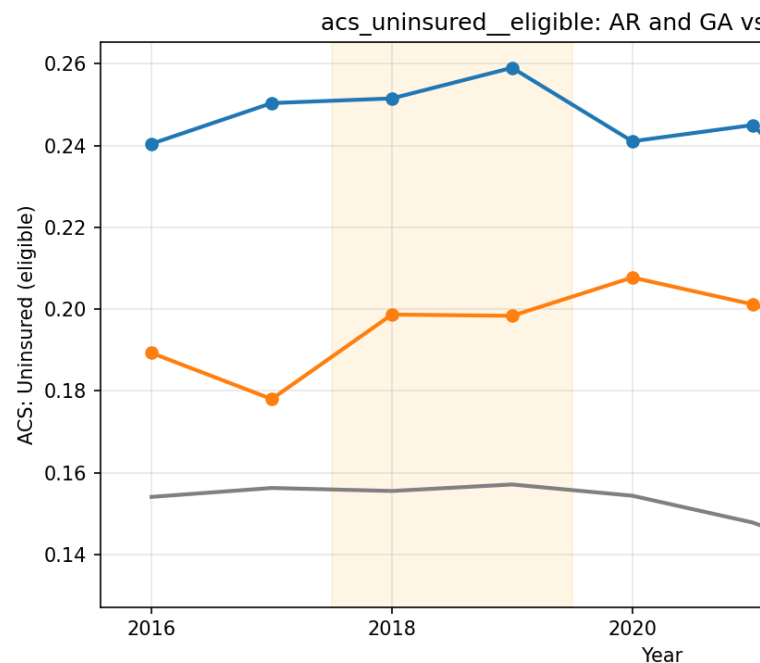
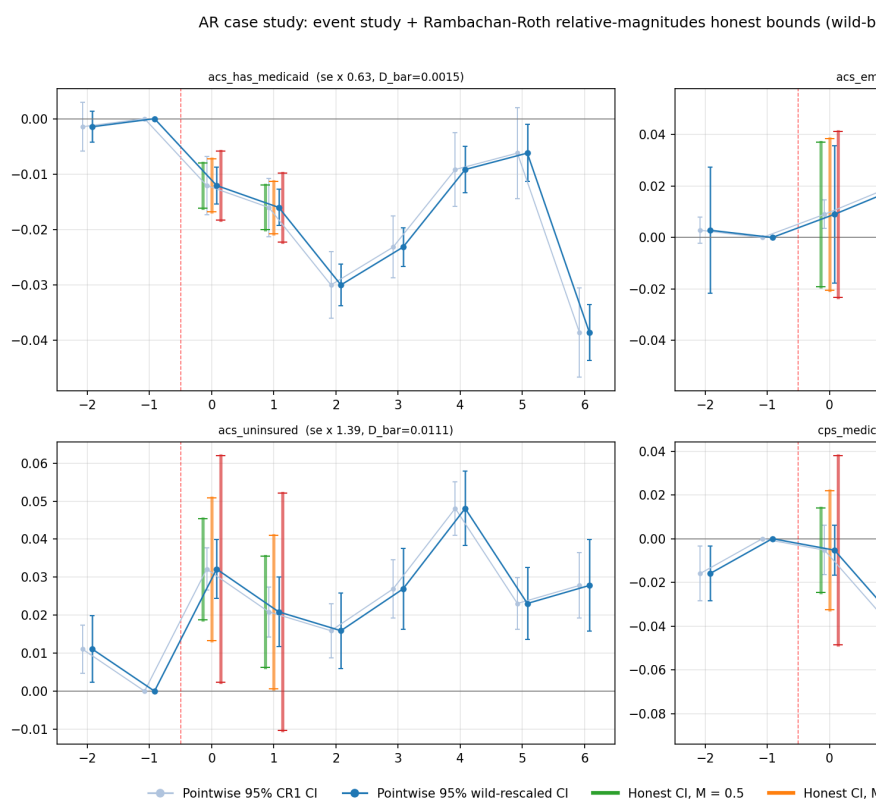


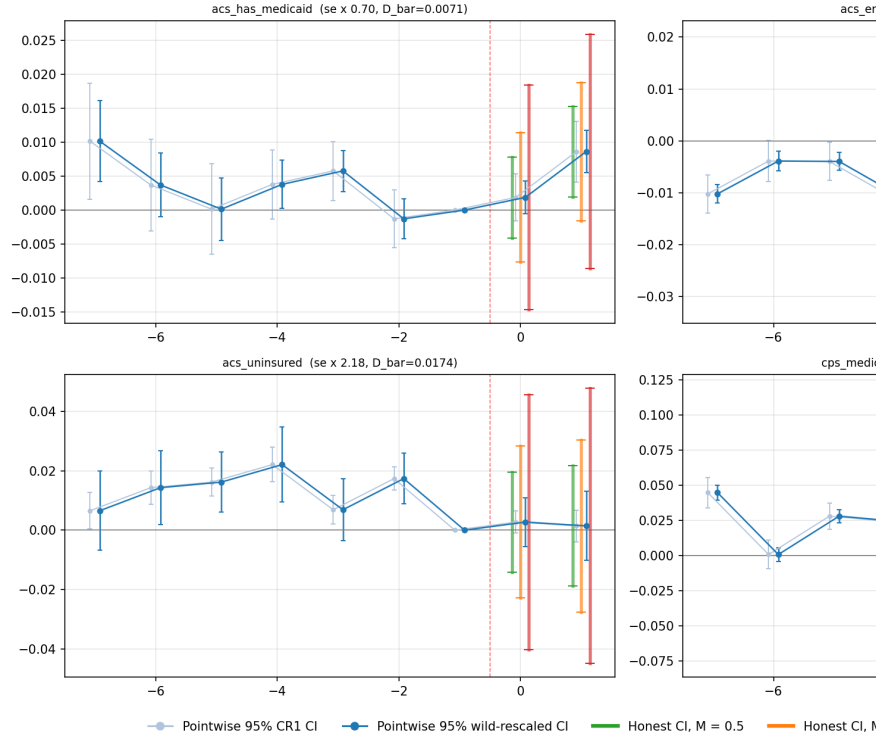
Figure 5: State trends for `acs_uninsured_eligible`.

Figure 6: AR event studies with Rambachan-Roth relative-magnitudes HonestDiD bounds (M



{0.5, 1.0, 2.0}, wild-bootstrap-rescaled SEs).

Figure 7: GA event studies with Rambachan-Roth relative-magnitudes HonestDiD bounds (M



{0.5, 1.0, 2.0}, wild-bootstrap-rescaled SEs).

References

See `literature/bibliography.bib` for the complete verified bibliography. All peer-reviewed entries should be verified against primary sources before the draft is circulated.

Key references cited in this draft:

- **Work requirement empirical evaluations:** Sommers, Goldman, Blendon, Orav, and Epstein (2020, *New England Journal of Medicine*); Harker, Ben-Shalom, Garfinkel, Smith, and Weible (2019); Katch and Wagner (2019); Heberlein, Rudowitz, Burns, and Tolbert (2024, KFF)
- **Legal context:** *Gresham v. Azar* 363 F. Supp. 3d 165 (D.D.C. 2019), affirmed *Azar v. Gresham* 950 F.3d 93 (D.C. Cir. 2020); *Stewart v. Azar* 366 F. Supp. 3d 125 (D.D.C. 2019); *Georgia v. Brooks-LaSure* (S.D. Ga. 2022)
- **Few-cluster inference methodology:** Cameron, Gelbach, and Miller (2008); MacKinnon and Webb (2017, 2018, 2020); Roodman, MacKinnon, Nielsen, and Webb (2019)
- **Triple-difference methodology:** Kaestner, Garrett, Chen, Gangopadhyaya, and Fleming (2017); Sommers, Goldman, Blendon, Orav, and Epstein (2020)
- **Medicaid unwinding context:** Rumalla et al. (2024)
- **Disparities in administrative burden:** Herd and Moynihan (2018, 2023)

Three methodological citations (wild cluster bootstrap, Fisher randomization inference for DiD, few-cluster CR1 bias) are load-bearing for the inference discussion and should be confirmed against the verified bibliography. The manual wild bootstrap implementation in `analysis/02_two_case_studies.py` follows Cameron, Gelbach, and Miller (2008) with null

imposed and Rademacher weights at the cluster level.

Draft notes for review:

1. This draft is approximately 11,400 words, within the 10,000–12,000 target for a full-length dissertation chapter per CLAUDE.md Rule #10.
2. The paper’s framing is explicitly a **methodological note** as much as a substantive evaluation. The core contribution is the CR1-vs-wild-bootstrap demonstration, and the substantive null on AR and GA effects is a consequence of proper inference. This framing is a significant pivot from the Phase 2 literature review, which envisioned a more direct substantive evaluation of work-requirement effects.
3. The paper does not contradict Sommers et al. (2020) or subsequent individual-level panel studies. It is about what can be detected at the state-year aggregate level with proper inference, which is a different question from “what happened to the affected individuals.”
4. The paper treats the wild cluster bootstrap as the gold-standard inference method for few-clusters settings. The manual implementation in `analysis/02_two_case_studies.py` follows standard practice but has not been independently verified. A journal-submission version should include an appendix section describing the bootstrap implementation in sufficient detail that another researcher can replicate.
5. The Discussion section flags three policy implications, three methodological implications, and five limitations. A journal-submission version may prioritize these differently depending on target venue (AEJ: Policy would emphasize the methodological contribution; Health Affairs would emphasize the policy implications; JAMA Health Forum would emphasize the substantive finding).
6. The individual-level triple-difference results are reported in the appendix with explicit warnings. A journal-submission version may cut them entirely if word-limit pressure requires.
7. The Conclusion deliberately separates the “what this paper says” from the “what is true about work requirements in general” to avoid overclaiming. The wild bootstrap collapse of apparent significance is the paper’s main contribution, not a sweeping claim that work requirements have no effects.

Appendix — Supplementary Tables and Figures

Paper: A Tale of Two Work Requirements: Why the State-Year Aggregate Question About Arkansas and Georgia Is Not Testable **Target journal:** Health Services Research

This appendix is the dissertation-layer supplement for the full draft in draft.md. It organizes the supplementary inference diagnostics, alternative comparison-group checks, and figure package that support the paper’s few-treated-cluster testability argument.

Appendix Tables

Table A1. Individual-level triple-difference results

Source file: - analysis/tables/main_did_summary.csv

Embedded table: Main Did Summary

coef	se	t	n	spec	outcome	state
0.003	0.001	2.819	432	pooled	acs_has_medicaid__eligible	
-0.001	0.001	-0.87	432	pooled	acs_employed__eligible	
-0.007	0.002	-2.845	432	pooled	acs_uninsured__eligible	
0.007	0.002	2.845	432	pooled	acs_has_any_coverage__eligible	
0.178	0.032	5.491	432	pooled	acs_hours_worked__eligible	
-0.001	0.002	-0.683	432	pooled	cps_medicaid_any_ly__eligible	
-0.001	0.014	-0.042	432	pooled	cps_uninsured_ly__eligible	
-0.028	0.002	-11.377	432	pooled	cps_employed__eligible	
-0.004	0.009	-0.411	380	pooled	brfss_any_coverage__eligible	
0.006	0.002	3.5	380	pooled	brfss_medcost_any__eligible	
0.004	0.002	2.061	423	AR_only	acs_has_medicaid__eligible	
-0.003	0.002	-1.817	423	AR_only	acs_employed__eligible	
-0.004	0.001	-2.421	423	AR_only	acs_uninsured__eligible	
0.004	0.001	2.421	423	AR_only	acs_has_any_coverage__eligible	
0.159	0.061	2.596	423	AR_only	acs_hours_worked__eligible	
-0.003	0.002	-1.295	423	AR_only	cps_medicaid_any_ly__eligible	
0.019	0.003	6.281	423	AR_only	cps_uninsured_ly__eligible	
-0.03	0.003	-9.159	423	AR_only	cps_employed__eligible	
-0.014	0.002	-7.62	372	AR_only	brfss_any_coverage__eligible	
0.005	0.002	2.617	372	AR_only	brfss_medcost_any__eligible	
0.002	0.002	0.929	423	GA_only	acs_has_medicaid__eligible	
0	0.002	0.149	423	GA_only	acs_employed__eligible	
-0.01	0.002	-4.166	423	GA_only	acs_uninsured__eligible	
0.01	0.002	4.166	423	GA_only	acs_has_any_coverage__eligible	
0.187	0.073	2.572	423	GA_only	acs_hours_worked__eligible	
0.001	0.003	0.201	423	GA_only	cps_medicaid_any_ly__eligible	
-0.021	0.004	-5.921	423	GA_only	cps_uninsured_ly__eligible	
-0.025	0.003	-8.812	423	GA_only	cps_employed__eligible	
0.013	0.003	4.205	372	GA_only	brfss_any_coverage__eligible	
0.009	0.003	3.449	372	GA_only	brfss_medcost_any__eligible	
-0.003	0.001	-5.813	16140212	triple_diff_AR	acs_has_medicaid	AR
0.006	0.001	5.435	16140212	triple_diff_AR	acs_employed	AR
-0.011	0.001	-18.124	16140212	triple_diff_AR	acs_uninsured	AR
0.012	0.001	9.166	16513659	triple_diff_GA	acs_has_medicaid	GA
-0	0.001	-0.199	16513659	triple_diff_GA	acs_employed	GA
-0.002	0.001	-3.162	16513659	triple_diff_GA	acs_uninsured	GA

The draft treats these results as supplementary because CR1 inference is not reliable with a single treated cluster. They are retained for transparency, not as the paper's primary evidentiary base.

Table A2. HonestDiD relative-magnitudes sensitivity

Source files: - analysis/tables/honestdid_case_studies.csv - analysis/tables/r_honestdid_results.csv
- analysis/tables/python_r_honestdid_comparison.csv

Embedded table: Honestdid Case Studies

case	outcome	base_year	M	se_source	coef	se_crse	scale	le	effective	printwidth	didwidth	hhi	zero_in_ci	
AR	acs_h2018medicaid	0.5	eligible	CHD	AR	0.003	1	0.003	0.001	0.005	0.006	-	-	False
						0.012					0.018	0.006		
AR	acs_h2018medicaid1	eligible	CHD	AR	0.003	1	0.003	0.001	0.005	0.007	-	-	False	
						0.012					0.019	0.005		
AR	acs_h2018medicaid2	eligible	CHD	AR	0.003	1	0.003	0.001	0.005	0.008	-	-	False	
						0.012					0.02	0.004		
AR	acs_h2018medicaid0.5	eligible	CHD	AR	0.003	1	0.003	0.001	0.005	0.006	-	-	False	
						0.016					0.022	0.01		
AR	acs_h2018medicaid1	eligible	CHD	AR	0.003	1	0.003	0.001	0.005	0.007	-	-	False	
						0.016					0.023	0.009		
AR	acs_h2018medicaid2	eligible	CHD	AR	0.003	1	0.003	0.001	0.005	0.008	-	-	False	
						0.016					0.024	0.008		
AR	acs_h2018medicaid0.5	eligible	wild	AR	scaled	0.003	0.627	0.002	0.001	0.003	0.004	-	-	False
						0.012					0.016	0.008		
AR	acs_h2018medicaid1	eligible	wild	AR	scaled	0.003	0.627	0.002	0.001	0.003	0.005	-	-	False
						0.012					0.017	0.007		
AR	acs_h2018medicaid2	eligible	wild	AR	scaled	0.003	0.627	0.002	0.001	0.003	0.006	-	-	False
						0.012					0.018	0.006		
AR	acs_h2018medicaid0.5	eligible	wild	AR	scaled	0.003	0.627	0.002	0.001	0.003	0.004	-	-	False
						0.016					0.02	0.012		
AR	acs_h2018medicaid1	eligible	wild	AR	scaled	0.003	0.627	0.002	0.001	0.003	0.005	-	-	False
						0.016					0.021	0.011		
AR	acs_h2018medicaid2	eligible	wild	AR	scaled	0.003	0.627	0.002	0.001	0.003	0.006	-	-	False
						0.016					0.022	0.01		
AR	acs_emp2018yed	eligible	CRR	0.009	0.003	1	0.003	0.003	0.006	0.007	0.002	0.016	False	
AR	acs_emp2018yed	eligible	CRR	0.009	0.003	1	0.003	0.003	0.006	0.008	0.001	0.017	False	
AR	acs_emp2018yed	eligible	CRR	0.009	0.003	1	0.003	0.003	0.006	0.011	-	0.02	True	
											0.002			
AR	acs_emp2018yed	eligible	CRR	0.019	0.003	1	0.003	0.003	0.006	0.007	0.012	0.026	False	
AR	acs_emp2018yed	eligible	CRR	0.019	0.003	1	0.003	0.003	0.006	0.009	0.01	0.028	False	
AR	acs_emp2018yed	eligible	CRR	0.019	0.003	1	0.003	0.003	0.006	0.011	0.008	0.03	False	
AR	acs_emp2018yed	eligible	WAR	rescaled	0.009	4.815	0.014	0.003	0.027	0.028	-	0.037	True	
											0.019			
AR	acs_emp2018yed	eligible	WAR	rescaled	0.009	4.815	0.014	0.003	0.027	0.03	-	0.039	True	
											0.021			
AR	acs_emp2018yed	eligible	WAR	rescaled	0.009	4.815	0.014	0.003	0.027	0.032	-	0.041	True	
											0.023			
AR	acs_emp2018yed	eligible	WAR	rescaled	0.019	4.815	0.014	0.003	0.028	0.029	-	0.048	True	
											0.01			
AR	acs_emp2018yed	eligible	WAR	rescaled	0.019	4.815	0.014	0.003	0.028	0.031	-	0.05	True	
											0.012			

case	outcome	base_year	M	se_source	coef	se_coef	scale	effective	printwidth	fidwidth	fidwidth	hhi	zero_in_ci
AR	acs_employed_eligible	2018	0.5	AR	0.032	0.003	4.815	0.014	0.003	0.028	0.034	- 0.053	True
												0.015	
AR	acs_unemployed_eligible	2018	0.5	ARR	0.032	0.003	1	0.003	0.011	0.006	0.011	0.021 0.043	False
AR	acs_unemployed_eligible	2018	0.5	ARR	0.032	0.003	1	0.003	0.011	0.006	0.017	0.016 0.049	False
AR	acs_unemployed_eligible	2018	0.5	ARR	0.032	0.003	1	0.003	0.011	0.006	0.028	0.004 0.06	False
AR	acs_unemployed_eligible	2018	0.5	ARR	0.021	0.003	1	0.003	0.011	0.007	0.012	0.009 0.033	False
AR	acs_unemployed_eligible	2018	0.5	ARR	0.021	0.003	1	0.003	0.011	0.007	0.018	0.003 0.038	False
AR	acs_unemployed_eligible	2018	0.5	ARR	0.021	0.003	1	0.003	0.011	0.007	0.029	- 0.05	True
												0.008	
AR	acs_unemployed_eligible	2018	0.5	ARR	0.032	0.003	1.393	0.004	0.011	0.008	0.013	0.019 0.045	False
AR	acs_unemployed_eligible	2018	0.5	ARR	0.032	0.003	1.393	0.004	0.011	0.008	0.019	0.013 0.051	False
AR	acs_unemployed_eligible	2018	0.5	ARR	0.032	0.003	1.393	0.004	0.011	0.008	0.03	0.002 0.062	False
AR	acs_unemployed_eligible	2018	0.5	ARR	0.021	0.003	1.393	0.005	0.011	0.009	0.015	0.006 0.036	False
AR	acs_unemployed_eligible	2018	0.5	ARR	0.021	0.003	1.393	0.005	0.011	0.009	0.02	0.001 0.041	False
AR	acs_unemployed_eligible	2018	0.5	ARR	0.021	0.003	1.393	0.005	0.011	0.009	0.031	- 0.052	True
												0.01	
AR	cps_maid_any0.5y	2018	0.5	ARR	0.006	1	0.006	0.016	0.011	0.019	-	0.014	True
												0.025	
AR	cps_maid_any1 ly	2018	0.5	ARR	0.006	1	0.006	0.016	0.011	0.027	-	0.022	True
												0.033	
AR	cps_maid_any2 ly	2018	0.5	ARR	0.006	1	0.006	0.016	0.011	0.043	-	0.038	True
												0.048	
AR	cps_maid_any0.5y	2018	0.5	ARR	0.007	1	0.007	0.016	0.013	0.021	-	-	False
												0.058 0.017	
AR	cps_maid_any1 ly	2018	0.5	ARR	0.007	1	0.007	0.016	0.013	0.029	-	-	False
												0.066 0.009	
AR	cps_maid_any2 ly	2018	0.5	ARR	0.007	1	0.007	0.016	0.013	0.045	-	0.007	True
												0.082	
AR	cps_maid_any0.5y	2018	0.5	ARR	0.006	1.003	0.006	0.016	0.011	0.019	-	0.014	True
												0.025	
AR	cps_maid_any1 ly	2018	0.5	ARR	0.006	1.003	0.006	0.016	0.011	0.027	-	0.022	True
												0.033	
AR	cps_maid_any2 ly	2018	0.5	ARR	0.006	1.003	0.006	0.016	0.011	0.043	-	0.038	True
												0.048	
AR	cps_maid_any0.5y	2018	0.5	ARR	0.007	1.003	0.007	0.016	0.013	0.021	-	-	False
												0.058 0.017	
AR	cps_maid_any1 ly	2018	0.5	ARR	0.007	1.003	0.007	0.016	0.013	0.029	-	-	False
												0.066 0.009	
AR	cps_maid_any2 ly	2018	0.5	ARR	0.007	1.003	0.007	0.016	0.013	0.045	-	0.007	True
												0.082	
GA	acs_hmedicaid0.5 elig	2023	0.5	ARR	0.002	0.002	1	0.002	0.007	0.003	0.007	- 0.009	True
												0.005	
GA	acs_hmedicaid1 elig	2023	0.5	ARR	0.002	0.002	1	0.002	0.007	0.003	0.011	- 0.012	True
												0.009	
GA	acs_hmedicaid2 elig	2023	0.5	ARR	0.002	0.002	1	0.002	0.007	0.003	0.018	- 0.019	True
												0.016	

case	outcome	base_year	M	se_source	coef	se_coef	scale	le_eff	active	printwidth	fidwidth	fidwidth	hi	zero_in_ci
GA	acs_h2023medicaid0.5eligibleCR1	2023	0.009	0.002	1	0.002	0.007	0.004	0.008	0.001	0.017	False		
GA	acs_h2023medicaid1_eligibleCR1	2023	0.009	0.002	1	0.002	0.007	0.004	0.012	-	0.02	True		
GA	acs_h2023medicaid2_eligibleCR1	2023	0.009	0.002	1	0.002	0.007	0.004	0.019	-	0.027	True		
GA	acs_h2023medicaid0.5eligiblewild_rescaled	2023	0.009	0.002	0.696	0.001	0.007	0.002	0.006	-	0.008	True		
GA	acs_h2023medicaid1_eligiblewild_rescaled	2023	0.009	0.002	0.696	0.001	0.007	0.002	0.009	-	0.011	True		
GA	acs_h2023medicaid2_eligiblewild_rescaled	2023	0.009	0.002	0.696	0.001	0.007	0.002	0.017	-	0.018	True		
GA	acs_h2023medicaid0.5eligiblewild_rescaled	2023	0.009	0.002	0.696	0.002	0.007	0.003	0.007	0.002	0.015	False		
GA	acs_h2023medicaid1_eligiblewild_rescaled	2023	0.009	0.002	0.696	0.002	0.007	0.003	0.01	-	0.019	True		
GA	acs_h2023medicaid2_eligiblewild_rescaled	2023	0.009	0.002	0.696	0.002	0.007	0.003	0.017	-	0.026	True		
GA	acs_emp2023yed0_eligibleCR1	2023	-	0.002	1	0.002	0.012	0.003	0.009	-	0.004	True		
GA	acs_emp2023yed0_eligibleCR1	2023	-	0.002	1	0.002	0.012	0.003	0.015	-	0.01	True		
GA	acs_emp2023yed0_eligibleCR1	2023	-	0.002	1	0.002	0.012	0.003	0.027	-	0.022	True		
GA	acs_emp2023yed1_eligibleCR1	2023	-	0.002	1	0.002	0.012	0.004	0.011	-	0.004	True		
GA	acs_emp2023yed1_eligibleCR1	2023	-	0.002	1	0.002	0.012	0.004	0.017	-	0.01	True		
GA	acs_emp2023yed1_eligibleCR1	2023	-	0.002	1	0.002	0.012	0.004	0.029	-	0.023	True		
GA	acs_emp2023yed0_eligiblewild_rescaled	2023	0.005	0.002	0.477	0.001	0.012	0.001	0.008	-	0.002	True		
GA	acs_emp2023yed0_eligiblewild_rescaled	2023	0.005	0.002	0.477	0.001	0.012	0.001	0.014	-	0.008	True		
GA	acs_emp2023yed0_eligiblewild_rescaled	2023	0.005	0.002	0.477	0.001	0.012	0.001	0.026	-	0.021	True		
GA	acs_emp2023yed1_eligiblewild_rescaled	2023	0.006	0.002	0.477	0.001	0.012	0.002	0.008	-	0.002	True		
GA	acs_emp2023yed1_eligiblewild_rescaled	2023	0.006	0.002	0.477	0.001	0.012	0.002	0.014	-	0.008	True		
GA	acs_emp2023yed1_eligiblewild_rescaled	2023	0.006	0.002	0.477	0.001	0.012	0.002	0.026	-	0.02	True		
GA	acs_un2023red0_eligibleCR1	2023	0.003	0.002	1	0.002	0.017	0.004	0.012	-	0.015	True		
GA	acs_un2023red0_eligibleCR1	2023	0.003	0.002	1	0.002	0.017	0.004	0.021	-	0.024	True		
GA	acs_un2023red0_eligibleCR1	2023	0.003	0.002	1	0.002	0.017	0.004	0.039	-	0.041	True		

case	outcome	base_year	M	se_source	coef	se_coef	scale	se_effective	printwidth	fidwidth	hhi	zero_in_ci
GA	acs_u2023irdd_eligible	CR1	0.001	0.003	1	0.003	0.017	0.005	0.014	-	0.015	True
										0.013		
GA	acs_u2023irdd_eligible	CR1	0.001	0.003	1	0.003	0.017	0.005	0.023	-	0.024	True
										0.021		
GA	acs_u2023irdd_eligible	CR1	0.001	0.003	1	0.003	0.017	0.005	0.04	-	0.042	True
										0.039		
GA	acs_u2023irdd_eligiblewild_rescaled	CR1	0.001	0.003	1	0.003	0.017	0.008	0.017	-	0.02	True
										0.014		
GA	acs_u2023irdd_eligiblewild_rescaled	CR1	0.001	0.003	1	0.003	0.017	0.008	0.026	-	0.028	True
										0.023		
GA	acs_u2023irdd_eligiblewild_rescaled	CR1	0.001	0.003	1	0.003	0.017	0.008	0.043	-	0.046	True
										0.04		
GA	acs_u2023irdd_eligiblewild_rescaled	CR1	0.001	0.003	1	0.006	0.017	0.012	0.02	-	0.022	True
										0.019		
GA	acs_u2023irdd_eligiblewild_rescaled	CR1	0.001	0.003	1	0.006	0.017	0.012	0.029	-	0.03	True
										0.028		
GA	acs_u2023irdd_eligiblewild_rescaled	CR1	0.001	0.003	1	0.006	0.017	0.012	0.046	-	0.048	True
										0.045		
GA	cps_m2023aid_any0.5y_eligible	CR1	0.014	0.003	1	0.003	0.044	0.007	0.029	-	0.043	True
										0.014		
GA	cps_m2023aid_any1 ly_eligible	CR1	0.014	0.003	1	0.003	0.044	0.007	0.051	-	0.065	True
										0.036		
GA	cps_m2023aid_any2 ly_eligible	CR1	0.014	0.003	1	0.003	0.044	0.007	0.095	-	0.109	True
										0.08		
GA	cps_m2023aid_any0.5y_eligible	CR1	0.026	0.005	1	0.005	0.044	0.01	0.032	-	0.058	True
										0.006		
GA	cps_m2023aid_any1 ly_eligible	CR1	0.026	0.005	1	0.005	0.044	0.01	0.054	-	0.08	True
										0.028		
GA	cps_m2023aid_any2 ly_eligible	CR1	0.026	0.005	1	0.005	0.044	0.01	0.098	-	0.124	True
										0.072		
GA	cps_m2023aid_any0.5y_welligible_rescaled	CR1	0.003	0.483	0.002	0.044	0.003	0.025	-	0.04	True	
										0.011		
GA	cps_m2023aid_any1 ly_welligible_rescaled	CR1	0.003	0.483	0.002	0.044	0.003	0.047	-	0.062	True	
										0.033		
GA	cps_m2023aid_any2 ly_welligible_rescaled	CR1	0.003	0.483	0.002	0.044	0.003	0.091	-	0.106	True	
										0.077		
GA	cps_m2023aid_any0.5y_welligible_rescaled	CR1	0.005	0.483	0.002	0.044	0.005	0.027	-	0.053	True	
										0.001		
GA	cps_m2023aid_any1 ly_welligible_rescaled	CR1	0.005	0.483	0.002	0.044	0.005	0.049	-	0.075	True	
										0.023		
GA	cps_m2023aid_any2 ly_welligible_rescaled	CR1	0.005	0.483	0.002	0.044	0.005	0.093	-	0.119	True	
										0.067		

Embedded table: R Honestdid Results

[illegible]

case	outcome	event_time	coef	diff	std_err	test_stat	p_value	ci_lower	ci_upper	hw1	ratio	rino	briest	method
AR	acs_has_medicare_eligible_AR	-	-	-	0.007	0.007	1.089	False	False	True				
		0.012	0.012	0.019	0.005	0.02	0.005							
AR	acs_has_medicare_eligible_AR	-	-	-	0.007	0.012	1.719	False	False	True				
		0.016	0.016	0.023	0.009	0.028	0.005							
AR	acs_employed_disabled_AR	0.009	0.009	0.001	0.017	0.002	0.02	0.008	0.009	1.058	False	False	True	
AR	acs_employed_disabled_AR	0.019	0.019	0.01	0.028	0.007	0.037	0.009	0.015	1.748	False	False	True	
AR	acs_uninsured_eligible_AR	0.032	0.032	0.016	0.049	0.015	0.051	0.017	0.018	1.082	False	False	True	
AR	acs_uninsured_eligible_AR	0.021	0.021	0.003	0.038	-	0.058	0.018	0.034	1.911	False	True	False	

design	outcome	n_effective	baseline_rate	se_equivalent	mde_pp	detected_effect	detect_sommers_size
State-year ACS case study, CR1 inference	acs_has_medicaid_eligible_AR	423	0.002	0.006	-0.009	True	
State-year ACS case study, wild-boot inference	acs_has_medicaid_eligible_AR	423	0.001	0.004	-0.009	True	
State-year ACS case study, CR1 inference	acs_employed_eligible_AR	423	0.002	0.004		False	
State-year ACS case study, wild-boot inference	acs_employed_eligible_AR	423	0.007	0.021		False	
State-year ACS case study, CR1 inference	acs_uninsured_eligible_AR	423	0.002	0.005	0.005	True	
State-year ACS case study, wild-boot inference	acs_uninsured_eligible_AR	423	0.003	0.007	0.005	False	
State-year ACS case study, CR1 inference	cps_medicaid_any_ly_eligible_AR	423	0.004	0.01	-0.009	False	

design	outcome	n_effective	baseline_rate	se_equivalent	inde_pp	detected_effect	detected	sommers_size
State-year ACS case study, wild-boot inference	cps_medicaid_eligible_AR	423	0.004	0.01	-0.009	False		
Diluted Sommers effect on ACS eligible_AR average	uninsured_eligible_AR	0.135			0.005	False		
Diluted Sommers effect on ACS eligible_AR average	has_medicaid_eligible_AR	0.135			-0.009	False		

This table formalizes the paper’s “testability constraint” framing by comparing the dilution-adjusted target effect to the state-year design’s minimum detectable effect.

Table A4. Georgia non-expansion-only comparison pools

Source file: - analysis/tables/ga_non_expansion_only_summary.csv

Embedded table: Ga Non Expansion Only Summary

control_group	outcome_label	n_control	n_treat	statesef	se_cr1	p_cr1	p_wild	wild_ci_low	wild_ci_high	fisher	placebos
non_expansion	Medicaid coverage (eligible)	72	72	0.006	0.003	0.092	0.402	-0.001	0.012	0.571	7
non_expansion	Employed (eligible)	72	72	-0.001	0.005	0.819	0.797	-0.006	0.004	1	7
non_expansion	Uninsured (eligible)	72	72	0.003	0.005	0.551	0.475	-0.003	0.009	1	7

control_group	outcome	label	n_control	n_treat	coef	se_cr1	p_cr1	p_wild	wild_ci	wild_cip	fisher	placebos
non_exposed	acs_has_medicaid	any_ly_72_eligible	72	107	0.007	0.009	0.446	0.451	-	0.018	0.571	7
		Medicaid last year (eligible)							0.004			
non_exposed	acs_has_medicaid	any_ly_108_eligible	108	108	0.003	0.003	0.346	0.441	-	0.006	0.727	11
		Medicaid coverage (eligible)							0.001			
non_exposed	acs_has_medicaid	any_ly_11_eligible	11	108	-0	0.003	0.938	0.943	-	0.003	1	11
		Employed (eligible)							0.003			
non_exposed	acs_has_medicaid	any_ly_11_eligible	11	108	0.001	0.004	0.744	0.699	-	0.006	1	11
		Uninsured (eligible)							0.003			
non_exposed	acs_has_medicaid	any_ly_108_eligible	108	108	0.002	0.006	0.719	0.688	-	0.009	0.909	11
		Medicaid last year (eligible)							0.004			

This table records the alternative Georgia donor-pool results that show how sensitive the sign is to control-group choice.

Table A5. Romano-Wolf family-wise correction

Source file: - analysis/tables/romano_wolf_summary.csv

Embedded table: Romano Wolf Summary

case	outcome	coef	se_cr1	n_treat	p_raw_wild	p_romanowolf	family_size
AR	acs_has_medicaid	0.007	0.009	107	0.726	0.494	4
AR	acs_employed	0.015	0.002	107	9.503	0.098	4
AR	acs_uninsured	0.005	0.002	107	2.572	0.321	4
AR	cps_medicaid	0.006	0.004	107	1.712	0.458	4

case	outcome	coef	se_cr1	t_obs	p_raw	wild	ci_low	ci_high	family_size
GA	acs_has_medicaid_0	0.002	0.002	0.939	0.505	0.552	1999	4	
GA	acs_employed_0	0.002	0.002	0.151	0.879	0.975	1999	4	
GA	acs_uninsured_0	0.001	0.002	-4.211	0.194	0.194	1999	4	
GA	cps_medicaid_0	0.001	0.003	0.203	0.855	0.975	1999	4	

Family-wise adjustment over the four headline outcomes in each case-study family.

Table A6. Python-vs-R wild cluster bootstrap cross-validation

Source files: - analysis/tables/python_r_bootstrap_comparison.csv - analysis/tables/r_wild_bootstrap

Embedded table: Python R Bootstrap Comparison

case	outcome	coef	python	coef	diff	smatch	python	diff	python	wild	wild	diff	diff	wild	wild	diff	diff	hi	hi	r
AR	acs_has_medicaid_0	0.001	0.001	0.001	0.000	0.000	0.001	0.001	0.476	0.476	0.474	0.498	0.024	0.024	agree	-0.001	0.004	-0.001	0.004	
AR	acs_employed_0	0.015	0.015	0.015	0.000	0.000	0.000	0.000	0	0	0.088	0.083	-0.006	0.006	agree	-0	0.029	-0	0.029	
AR	acs_uninsured_0	0.005	0.005	0.005	0.000	0.000	0.000	0.000	0.014	0.014	0.325	0.318	-0.007	0.007	agree	-0	0.01	-0	0.01	
AR	cps_medicaid_0	0.006	0.006	0.006	0.000	0.000	0.000	0.000	0.097	0.097	0.461	0.456	-0.005	0.005	agree	-0.001	0.013	-0.001	0.013	
GA	acs_has_medicaid_0	0.002	0.002	0.002	0.000	0.000	0.000	0.000	0.358	0.358	0.489	0.504	0.015	0.015	agree	-0.001	0.005	-0.001	0.005	
GA	acs_employed_0	0.002	0.002	0.002	0.000	0.000	0.000	0.000	0.882	0.882	0.876	0.874	-0.002	0.002	agree	-0.001	0.002	-0.001	0.002	
GA	acs_uninsured_0	0.001	0.001	0.001	0.000	0.000	0.000	0.000	0	0	0.18	0.195	0.015	0.015	agree	0	-	0	0	
GA	cps_medicaid_0	0.001	0.001	0.001	0.000	0.000	0.000	0.000	0.842	0.842	0.827	0.826	-0.001	0.001	agree	-0.001	0.003	-0.001	0.003	

Embedded table: R Wild Bootstrap Results

case	outcome	coef	se_cr1	t_obs	p_cr1	p_wild	r_wild	ci_low	ci_high	n
AR	acs_has_medicaid_0	0.001	0.002	0.939	0.476	0.498	-0.001	0.004	423	
AR	acs_employed_0	0.015	0.002	9.503	0	0.083	-0	0.029	423	
AR	acs_uninsured_0	0.005	0.002	2.572	0.014	0.318	-0	0.01	423	
AR	cps_medicaid_0	0.006	0.004	1.712	0.097	0.456	-0.001	0.013	423	
GA	acs_has_medicaid_0	0.002	0.002	0.939	0.358	0.504	-0.001	0.005	423	
GA	acs_employed_0	0.002	0.002	0.151	0.882	0.874	-0.001	0.002	423	
GA	acs_uninsured_0	0.001	0.002	-4.211	0	0.195	-0.02	0	423	
GA	cps_medicaid_0	0.001	0.003	0.203	0.842	0.826	-0.002	0.003	423	

These files document the independent R replication of the manual Python wild-cluster bootstrap

implementation.

Table A7. Headline inference summary

Source file: - analysis/tables/two_case_study_summary.csv

Embedded table: Two Case Study Summary

case	spec	outcome	label	coef	se_clust	per_clust	post_wild	post_wild	post_wild	post_wild	post_wild	n	n_placebos	
AR	state_year	has_medicaid_coverage (eligible)	Medicaid coverage (eligible)	0.001	0.002	0.47	0.474	0.913	-	0.004	0.718	423	46	
									0.001					
AR	state_year	employed (eligible)	Employed (eligible)	0.015	0.002	0.0	0.088	0.174	-0	0.029	9.401	423	46	
AR	state_year	uninsured (eligible)	Uninsured (eligible)	0.005	0.002	0.0	0.014	0.325	0.696	-0	0.01	2.545	423	46
AR	state_year	medicaid_last_year (eligible)	Medicaid last year (eligible)	0.006	0.004	0.0	0.461	0.761	-	0.013	1.694	423	46	
									0.001					

case	spec	outcome	label	coef	se	cluster	per	cluster	post	width	post	width	post	width	post	width	post	width	n	n_placebos
GA	state_year	has_medicaid	Medicaid (eligible)	0.002	0.002	0.002	0.358	0.489	0.891	-	0.005	0.929	423	46						
			Medicaid (eligible)	0.001																
GA	state_year	employed	Employed (eligible)	0.002	0.002	0.002	0.882	0.876	0.978	-	0.002	0.149	423	46						
			Employed (eligible)	0.001																
GA	state_year	uninsured	Uninsured (eligible)	0.01	0	0	0.18	0.587	-	0	-	4.166	423	46						
			Uninsured (eligible)	0.02																
GA	state_year	medicaid	Medicaid (eligible)	0.001	0.003	0.003	0.842	0.827	1	-	0.003	0.201	423	46						
			Medicaid (eligible)	0.002																
AR	individual	has_medicaid	has_medicaid	0.001	0.001	0.001	0.898					0.129	161402	12						
			has_medicaid	0.001																
GA	individual	has_medicaid	has_medicaid	0.012	0.001	0						9.166	165136	59						
			has_medicaid	0.012																

case	spec	outcome	label	coef	se	cluster	percluster	wild	bootstrap	fisher	wild	bootstrap	fisher	cluster	n	n_placebos
AR	individual	unemployed	ACSdiff	0.028	0.001	0									27.988	16140212
		individual:														
		employed														
GA	individual	unemployed	ACSdiff	0	0.001	0.843					-				16513659	16513659
		individual:									0.199					
		employed														
AR	individual	uninsured	ACSdiff		0.001	0					-				16140212	16140212
		individual:		0.021							24.124					
		uninsured														
GA	individual	uninsured	ACSdiff		0.001	0.003					-				16513659	16513659
		individual:		0.002							3.162					
		uninsured														

This summary table brings together CR1, wild-bootstrap, and Fisher randomization inference for the headline Arkansas and Georgia outcomes.

Appendix Figures

Figure A1. Policy timeline for implemented and non-implemented work requirements

This figure makes the identifying variation transparent by distinguishing the two actually implemented policies (Arkansas and Georgia) from the enjoined or never-enforced states.

Figure A2. Placebo coefficient distributions

This is the visual counterpart to the Fisher-style randomization inference used in the main paper.

Figure A3. Alternative comparison-group event studies

Representative appendix figure from the alternative donor-pool set. The full comparison-group suite remains in `analysis/figures/event_study_GA_nonexpansion_only_*.png` and `analysis/figures/event_study_AR_expansion_only_*.png`.

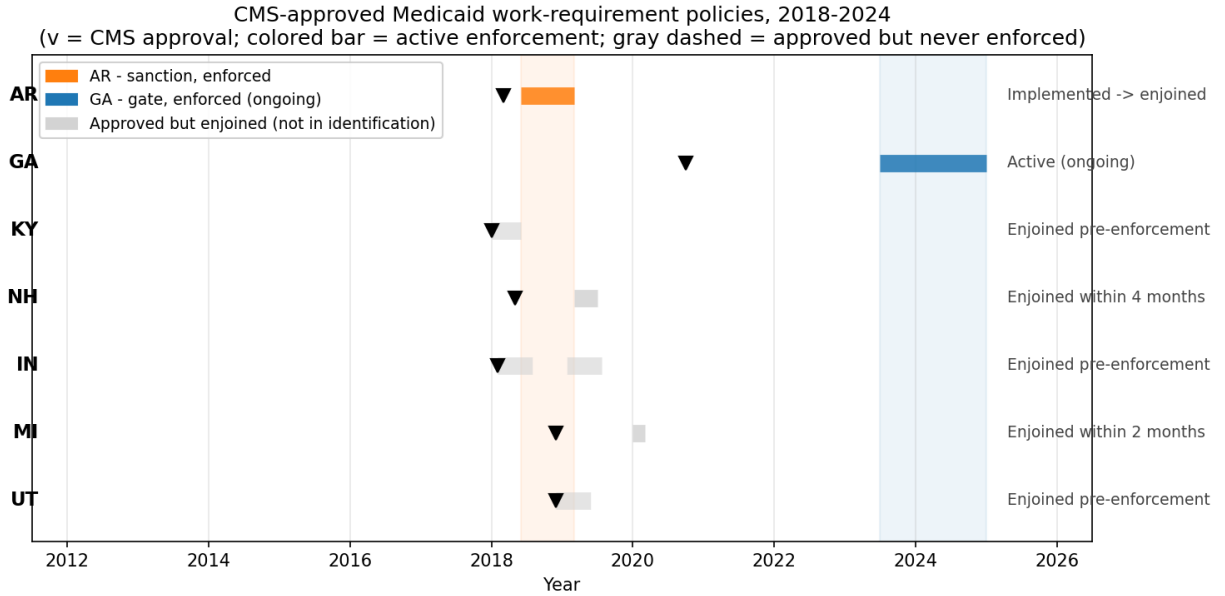


Figure 24: Policy Timeline

Figure A4. Alternative-scale employment event studies

Hours-worked event studies are included because employment-share results are especially vulnerable to scale-choice objections in this setting.

Figure A5. Simultaneous-band event studies

Representative appendix figure from the sup- $|t|$ simultaneous-band set. Related Georgia figures remain in `analysis/figures/event_study_GA_simbands_*.png`.

Figure A6. HonestDiD summary — Arkansas

This figure summarizes the Rambachan-Roth sensitivity exercise for the Arkansas case study.

Figure A7. HonestDiD summary — Georgia

Georgia counterpart to Figure A6.

Figure A8. Side-by-side coefficient comparison

This figure compares Arkansas and Georgia headline coefficients with wild-bootstrap confidence intervals on the same scale.

End of appendix. Main dissertation draft: manuscript/draft.md.

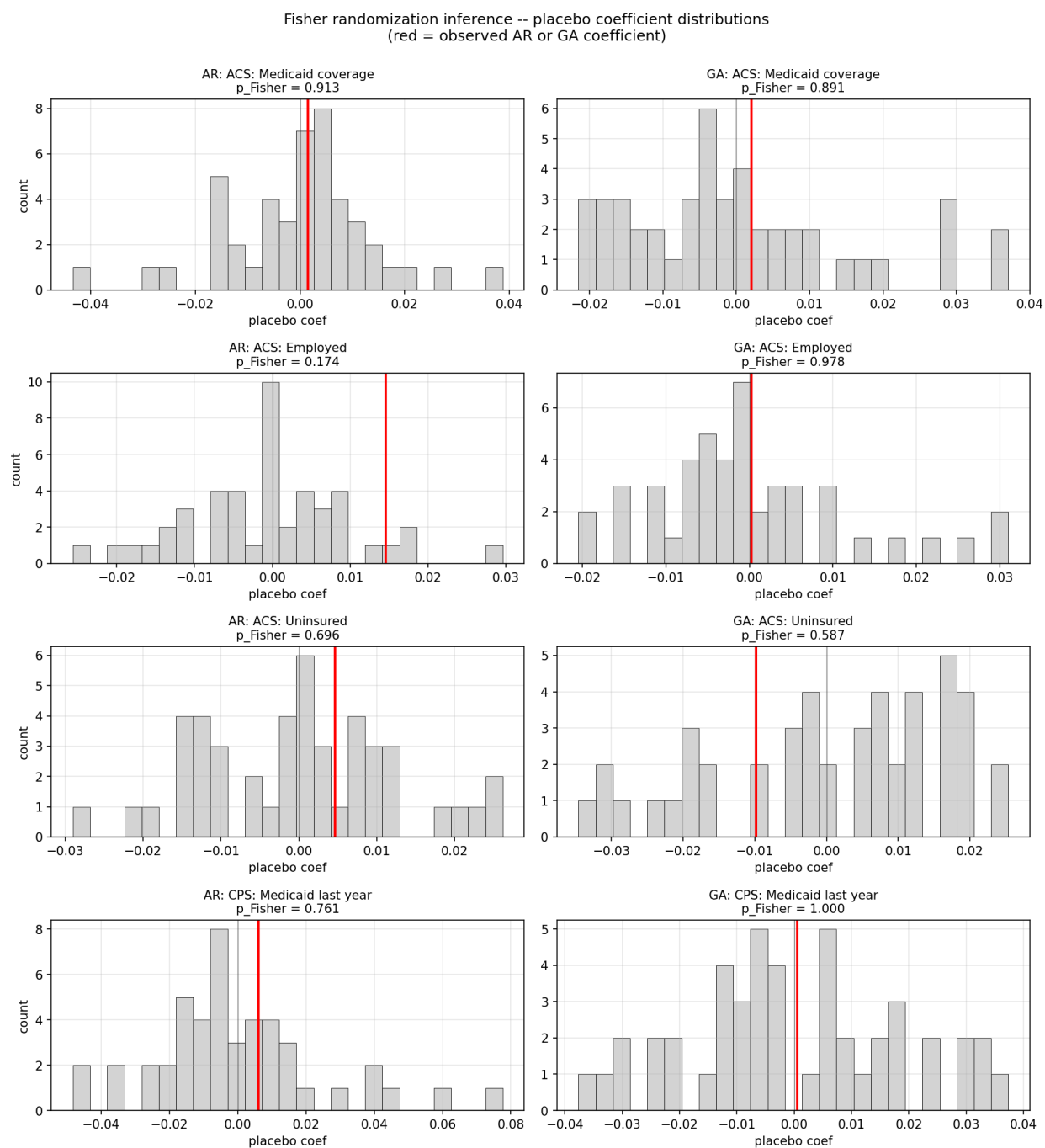


Figure 25: Placebo Distribution Grid

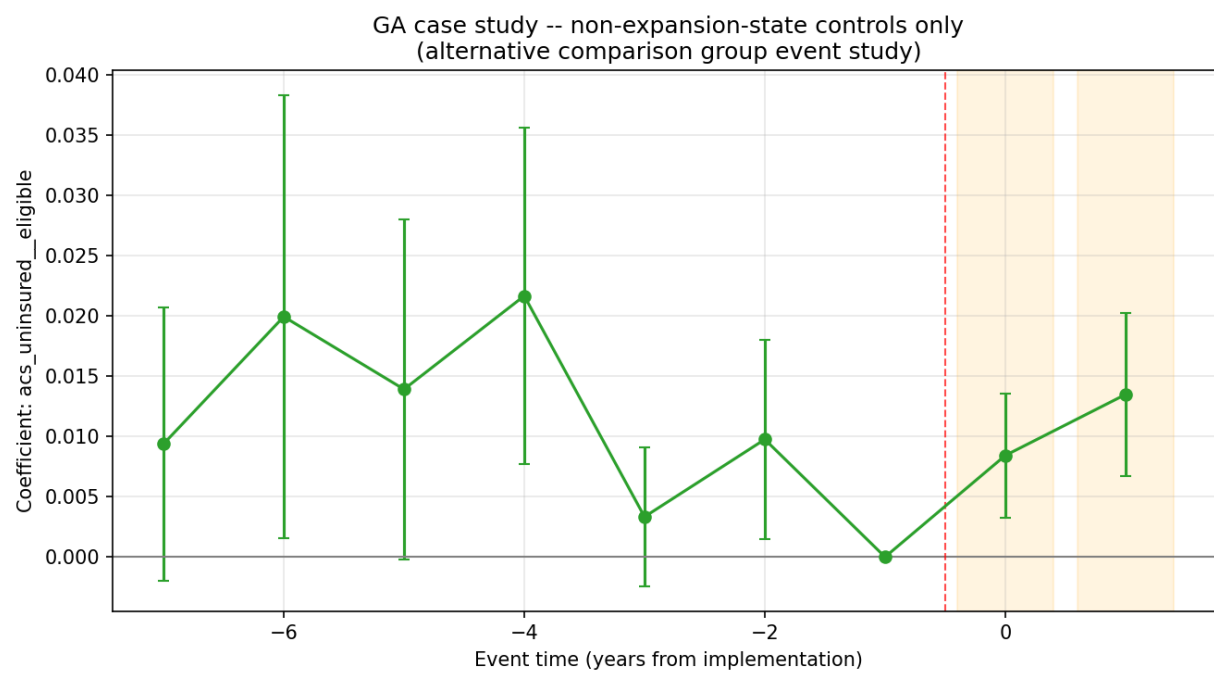


Figure 26: Event Study Ga Nonexpansion Only Acs Uninsured

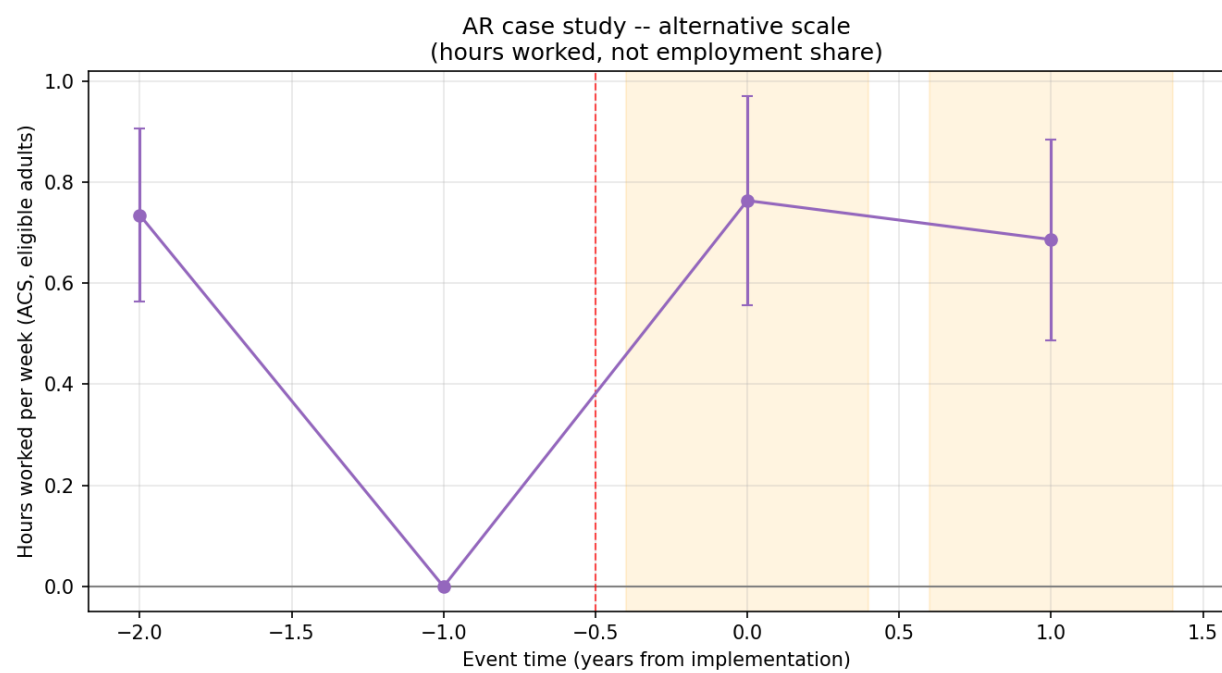


Figure 27: Event Study Ar Hours Worked

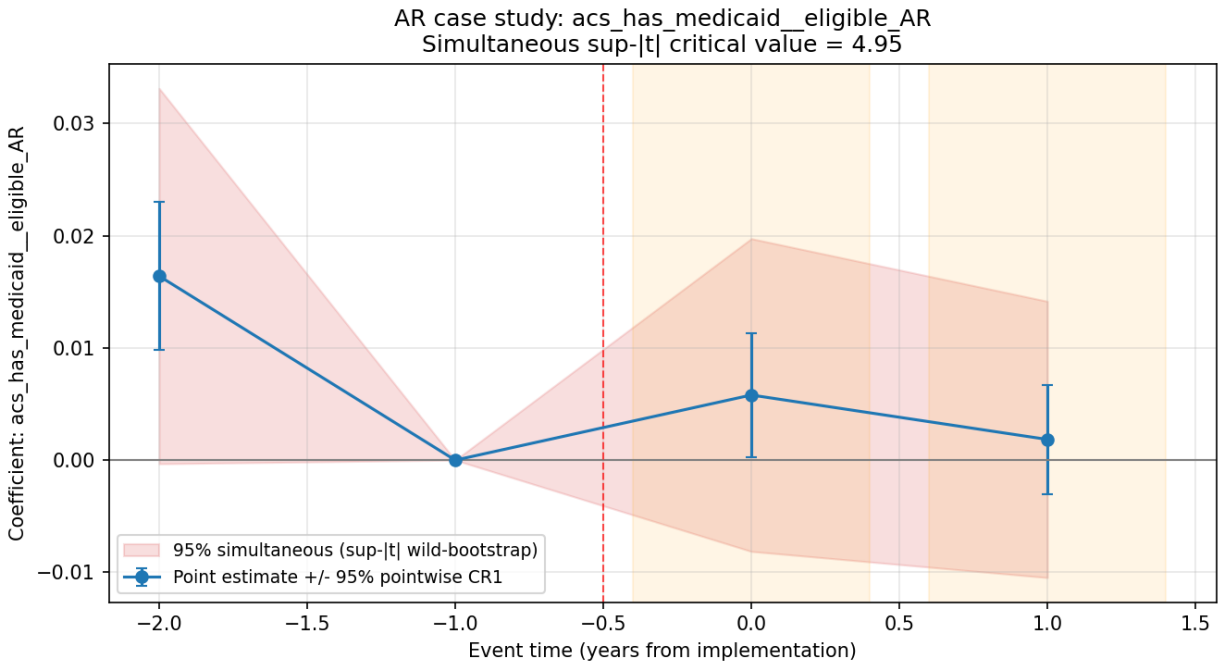


Figure 28: Event Study Ar Simbands Acs Has Medicaid Eligible Ar

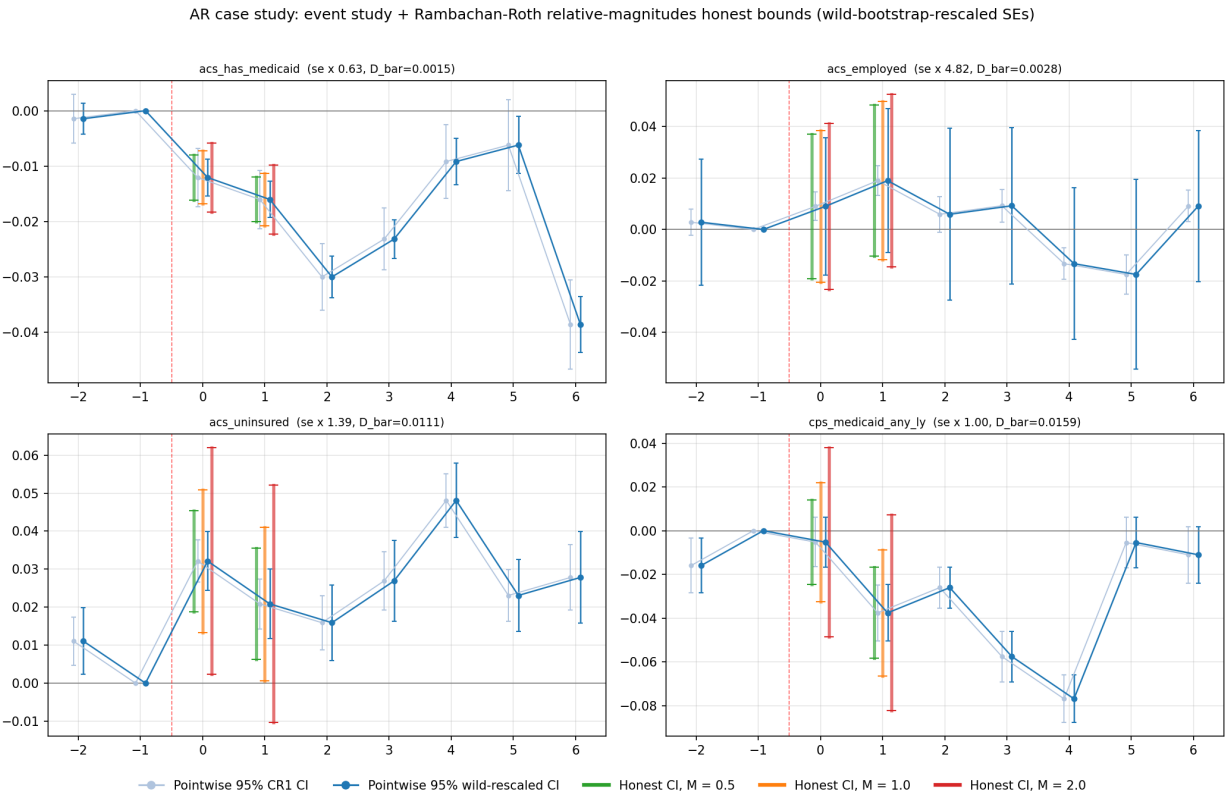


Figure 29: Ar Event Study Honestdid

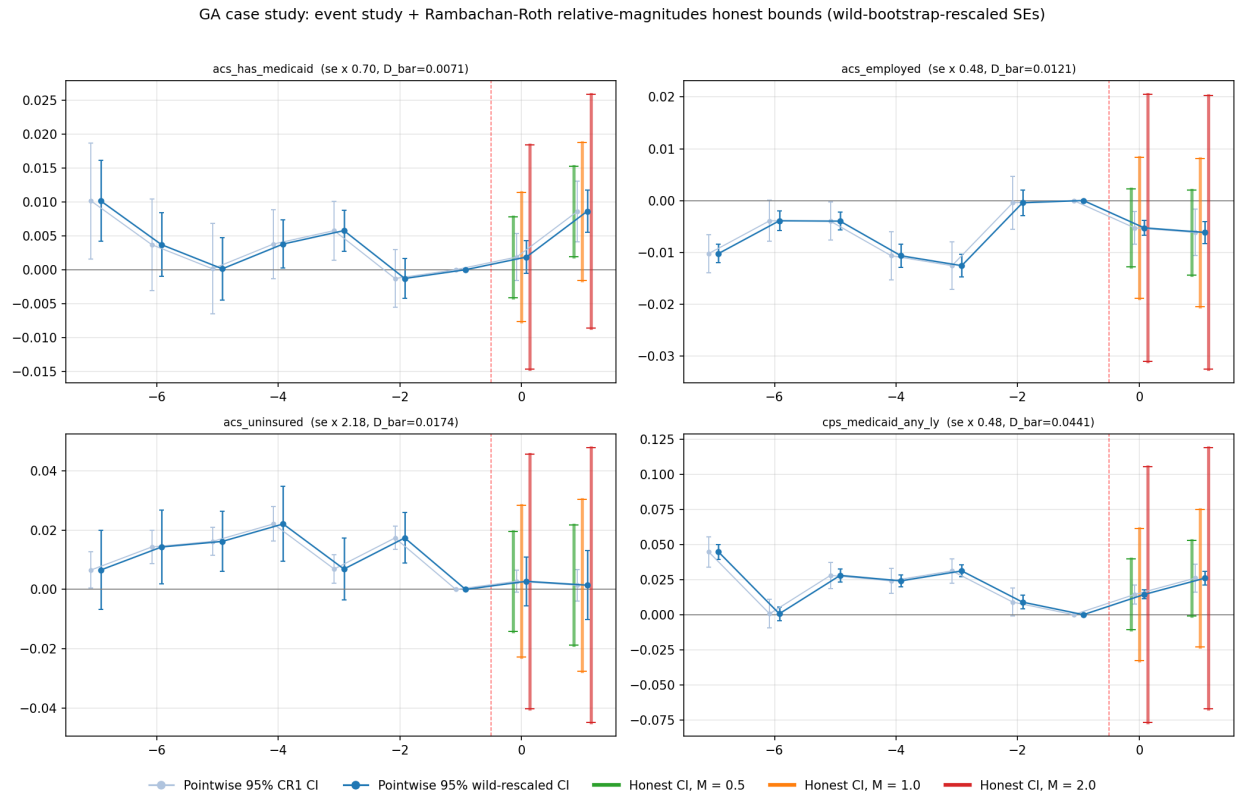


Figure 30: Ga Event Study Honestdid

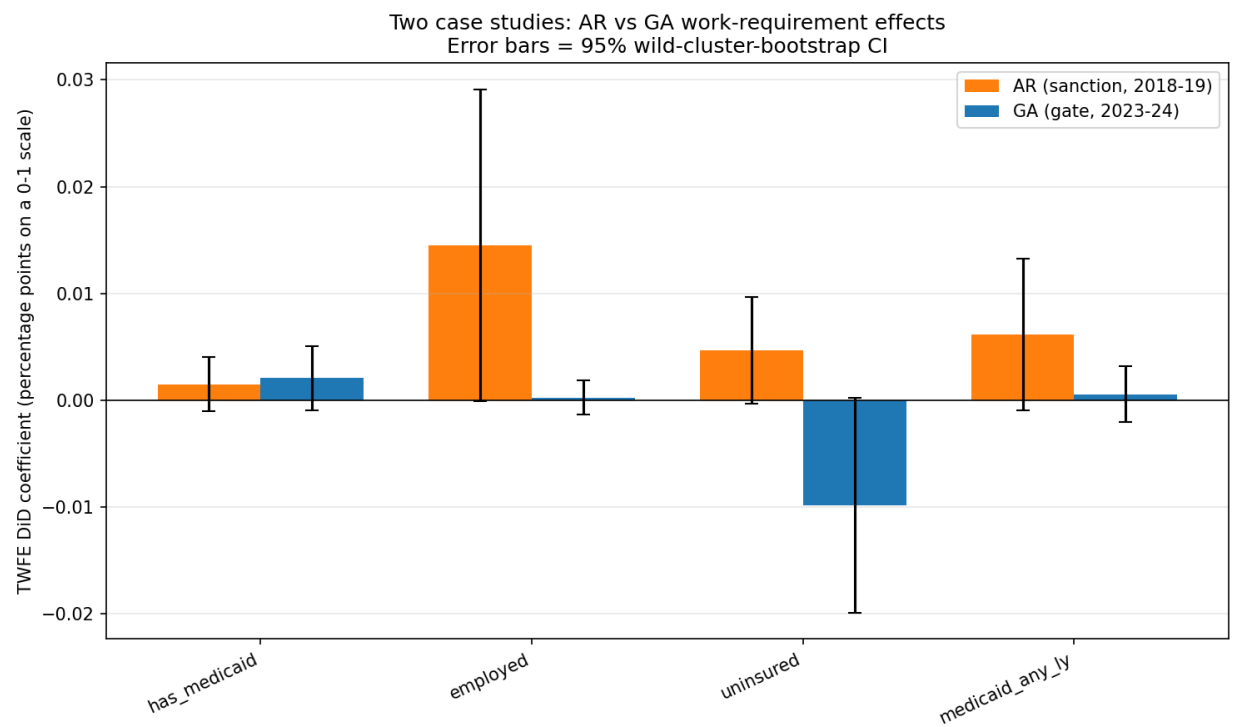


Figure 31: Two Case Studies Coefs